

REPORTE DE EJECUCIÓN
ALGORITMO GENÉTICO EN ESPACIO 2D Y 3D

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MÉTODOS UTILIZADOS:

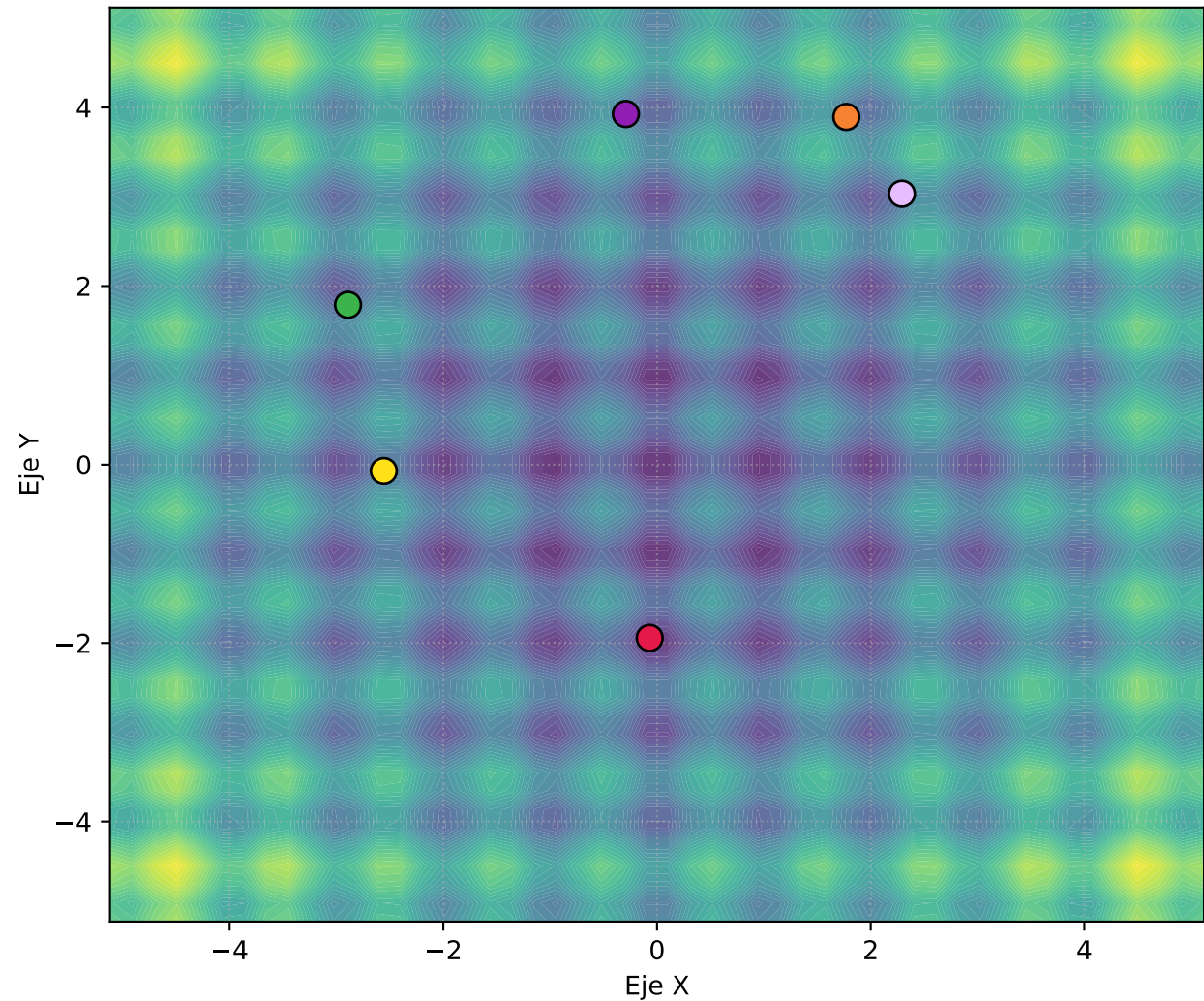
- > Aptitud 3D: □ RASTRIGIN
- > Codificación: □ BINARIO_2D
- > Selección: □ TORNEO
- > Cruza: □ UNIFORME
- > Mutación: □ MULTIPLE
- > Generaciones: □ 100

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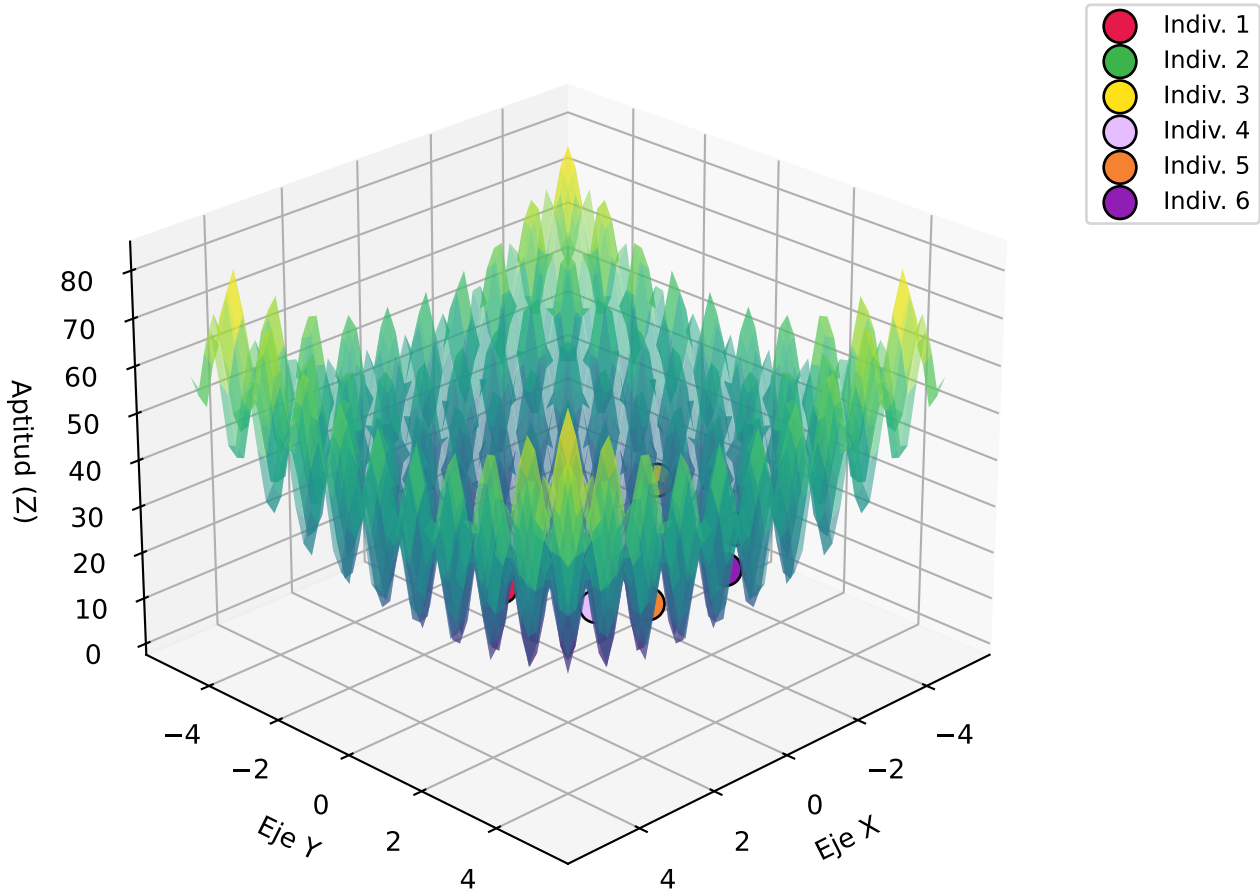
Centro de Investigaciones en Óptica

Generación 1: Posición Inicial de los 6 Mejores

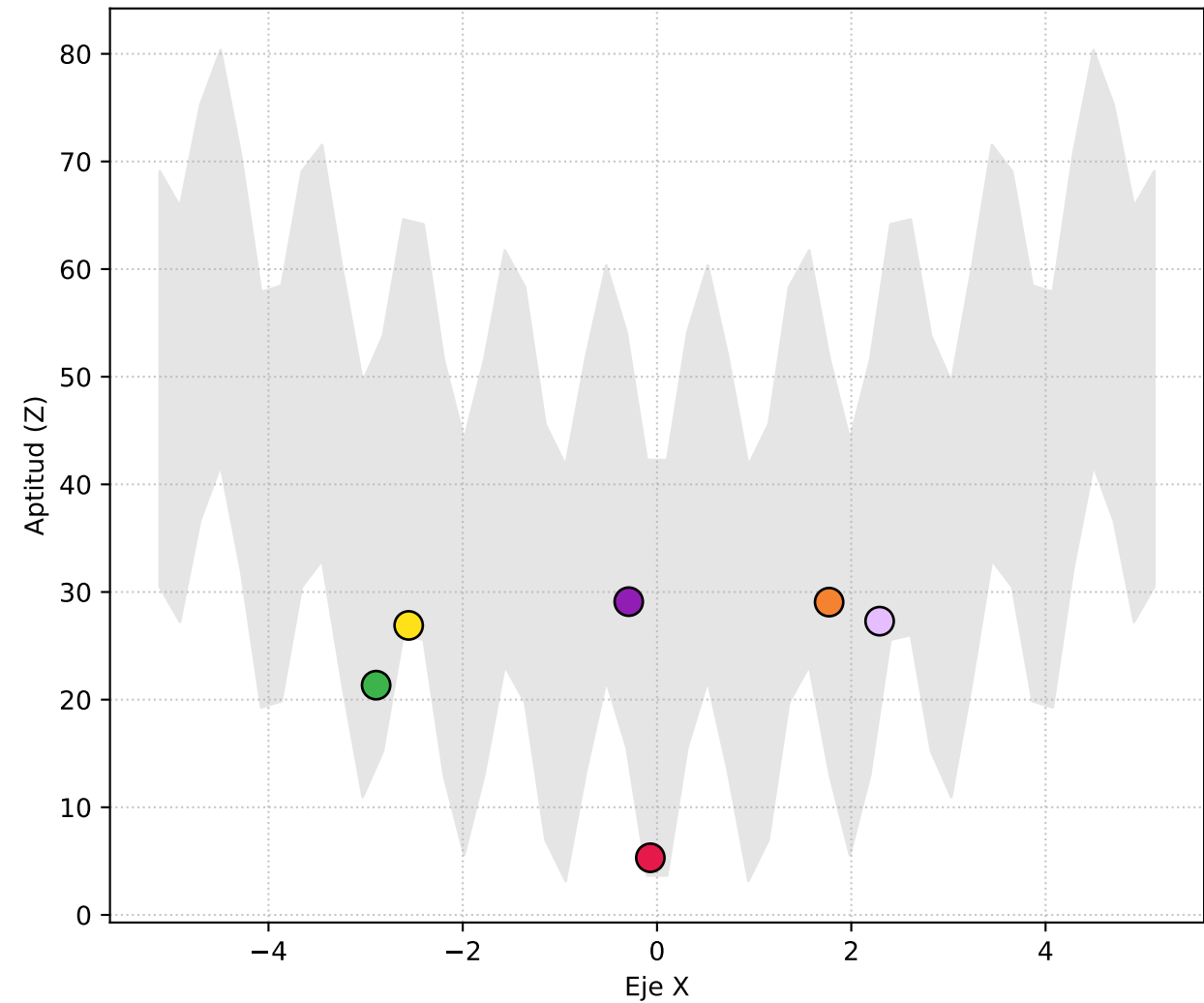
Vista Superior (X, Y)



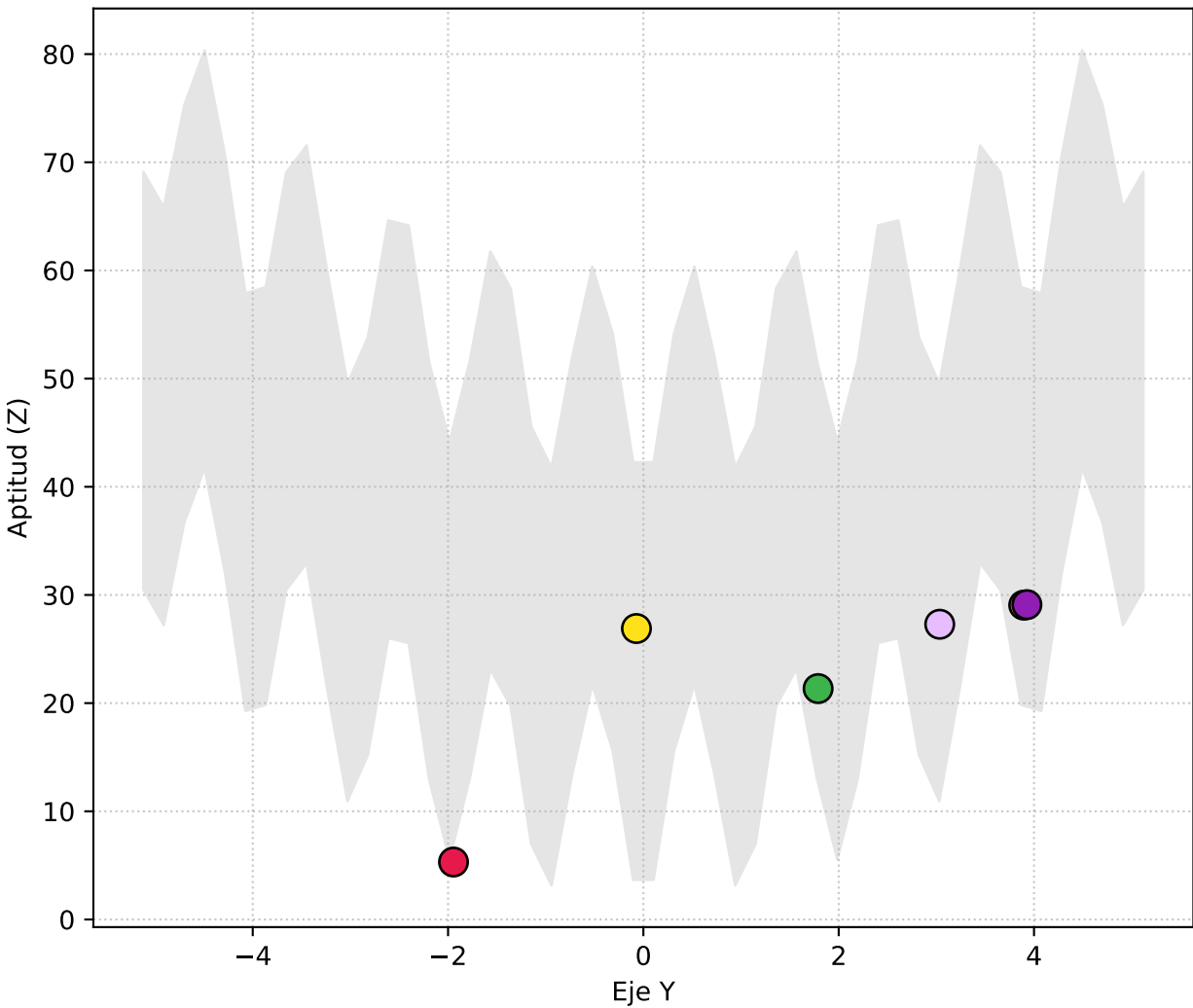
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

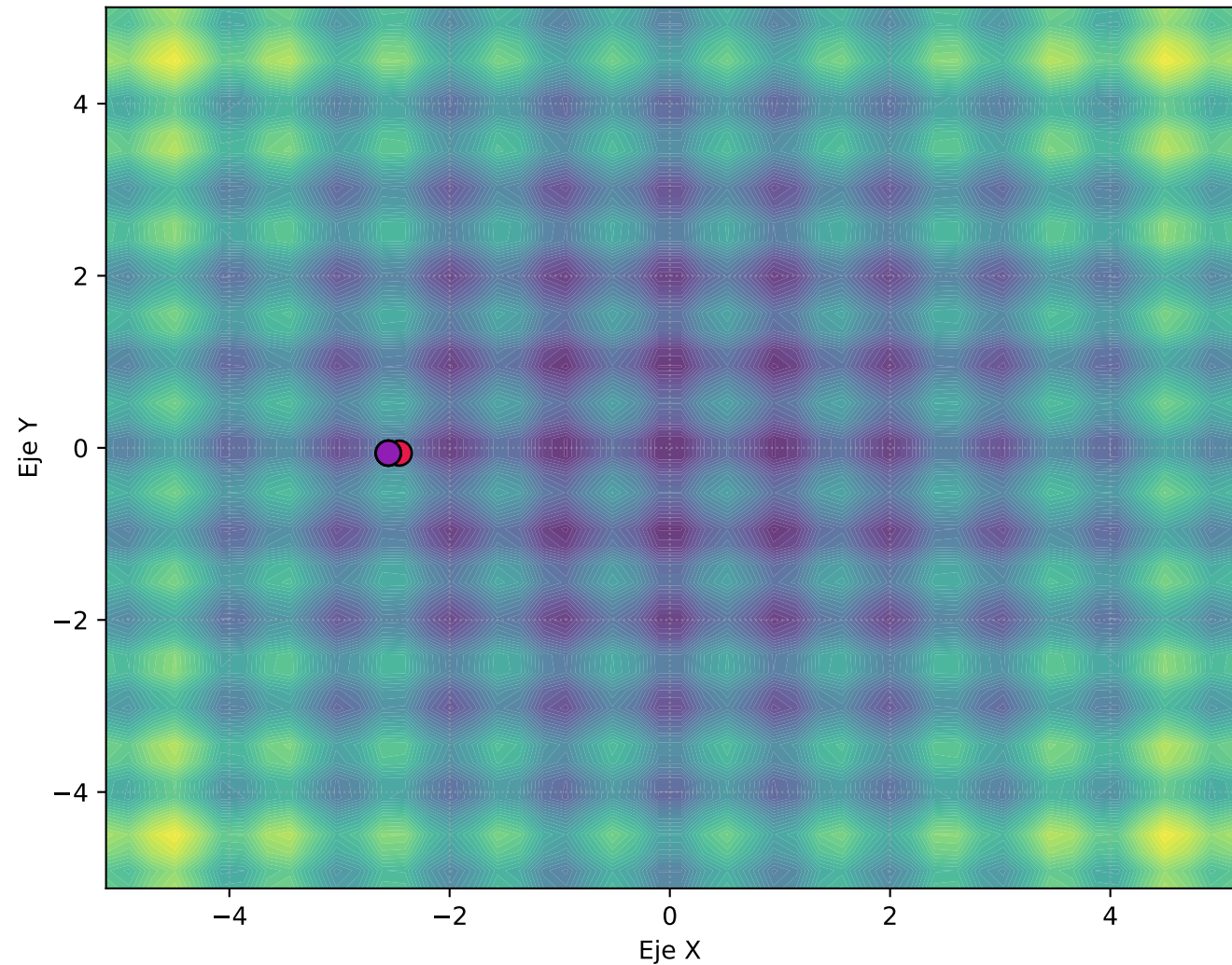


Vista Frontal (Y vs Elevación Z)

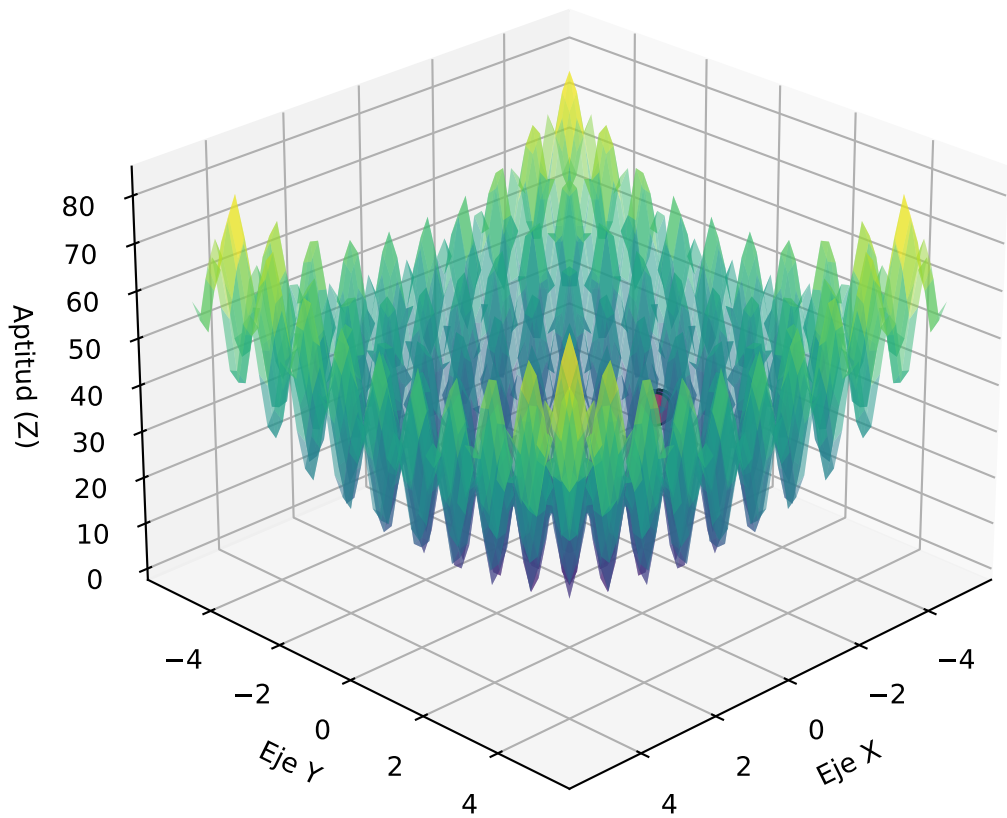


Progreso: Generación 10

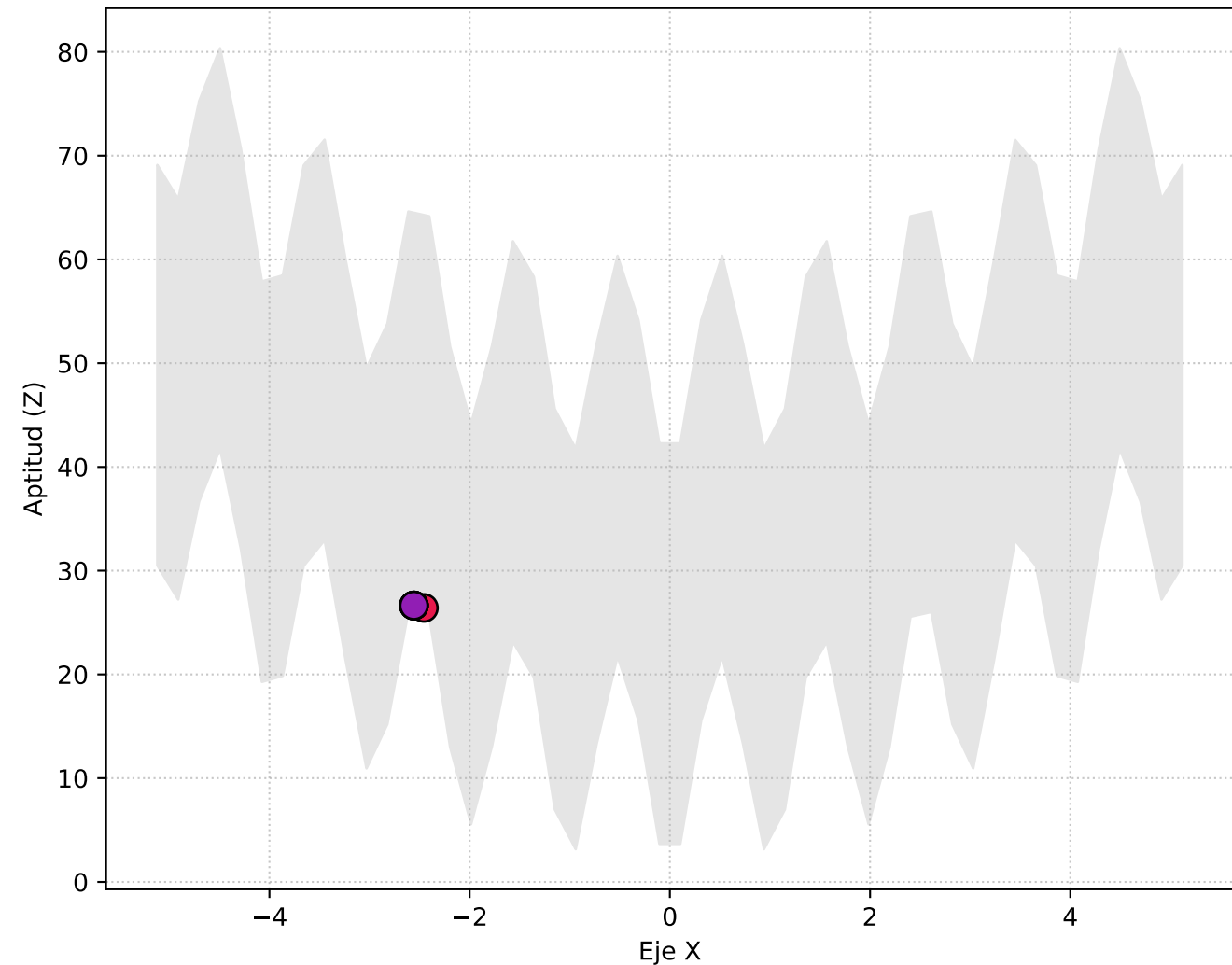
Vista Superior (X, Y)



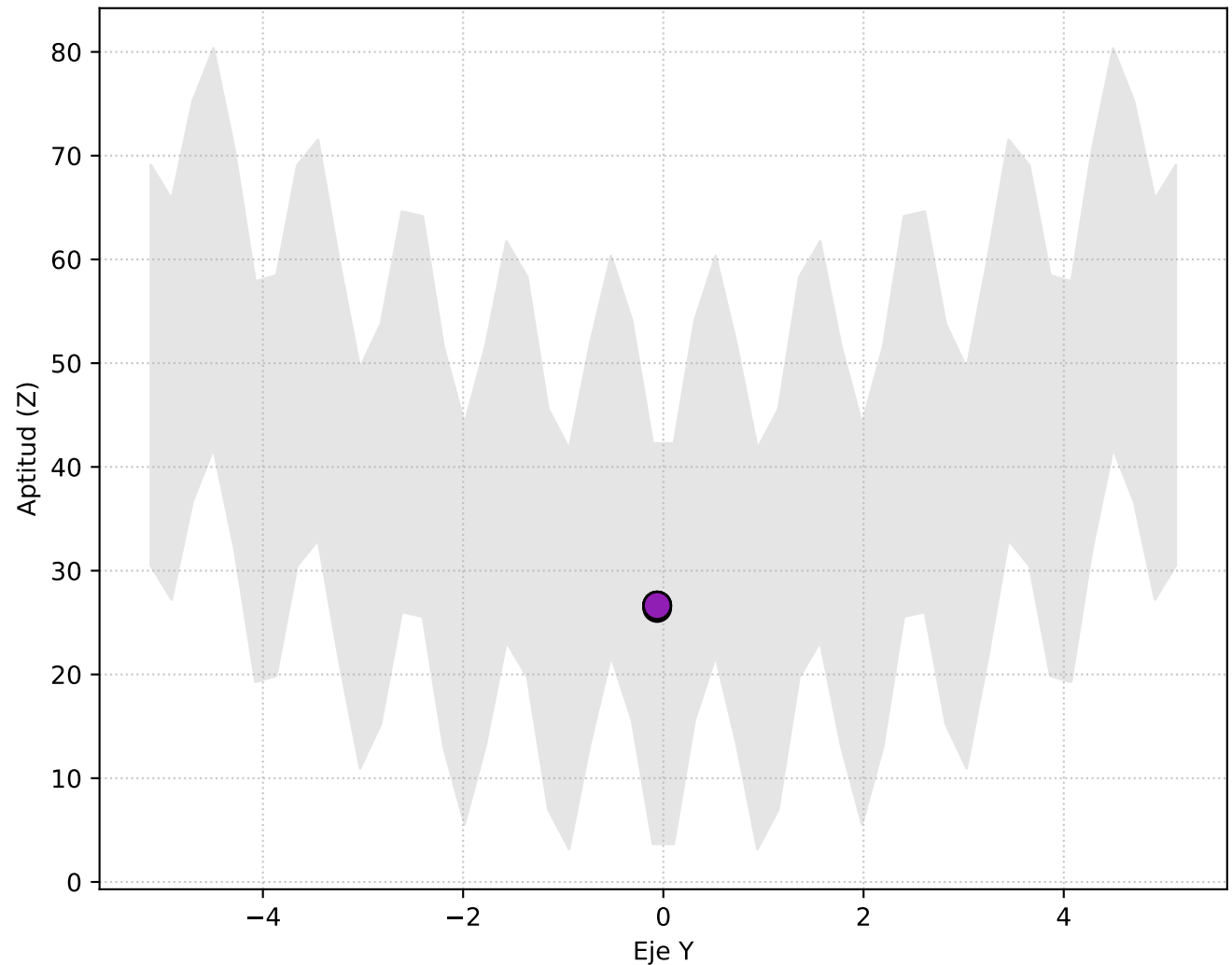
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

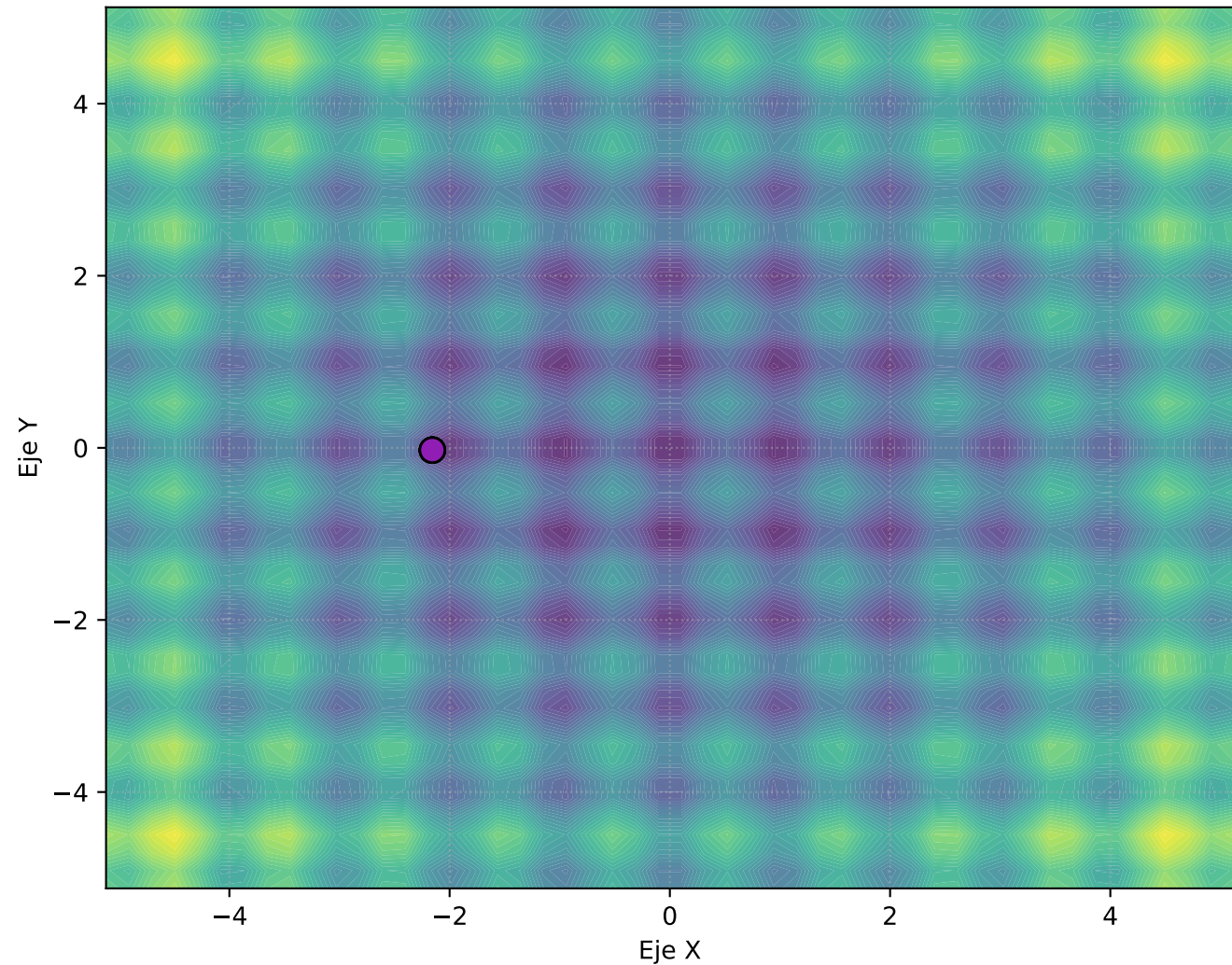


Vista Frontal (Y vs Elevación Z)

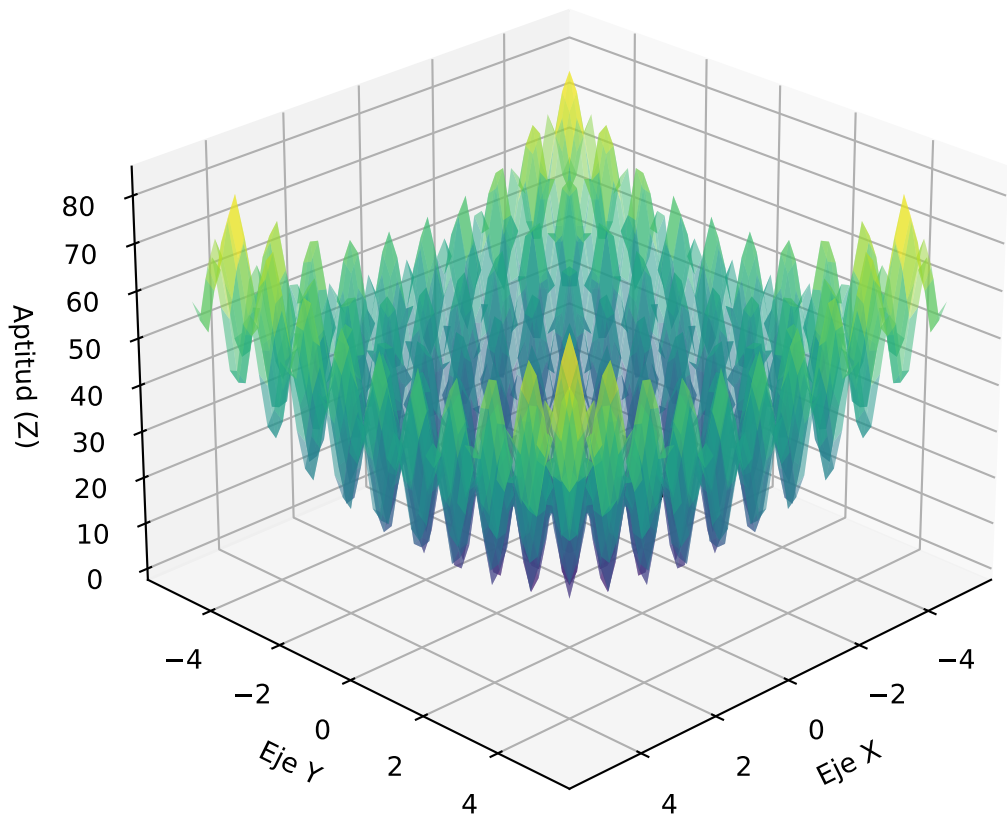


Progreso: Generación 20

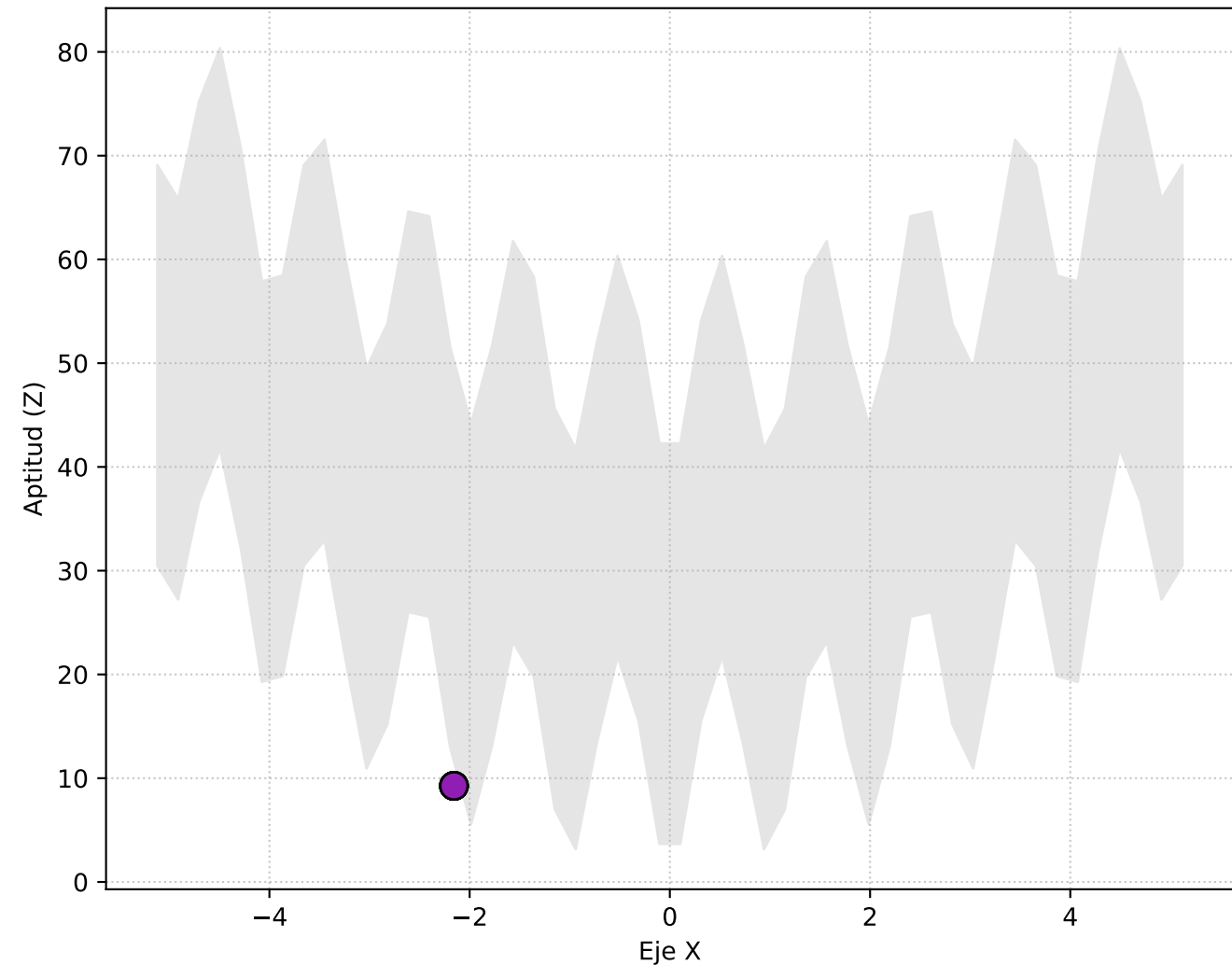
Vista Superior (X, Y)



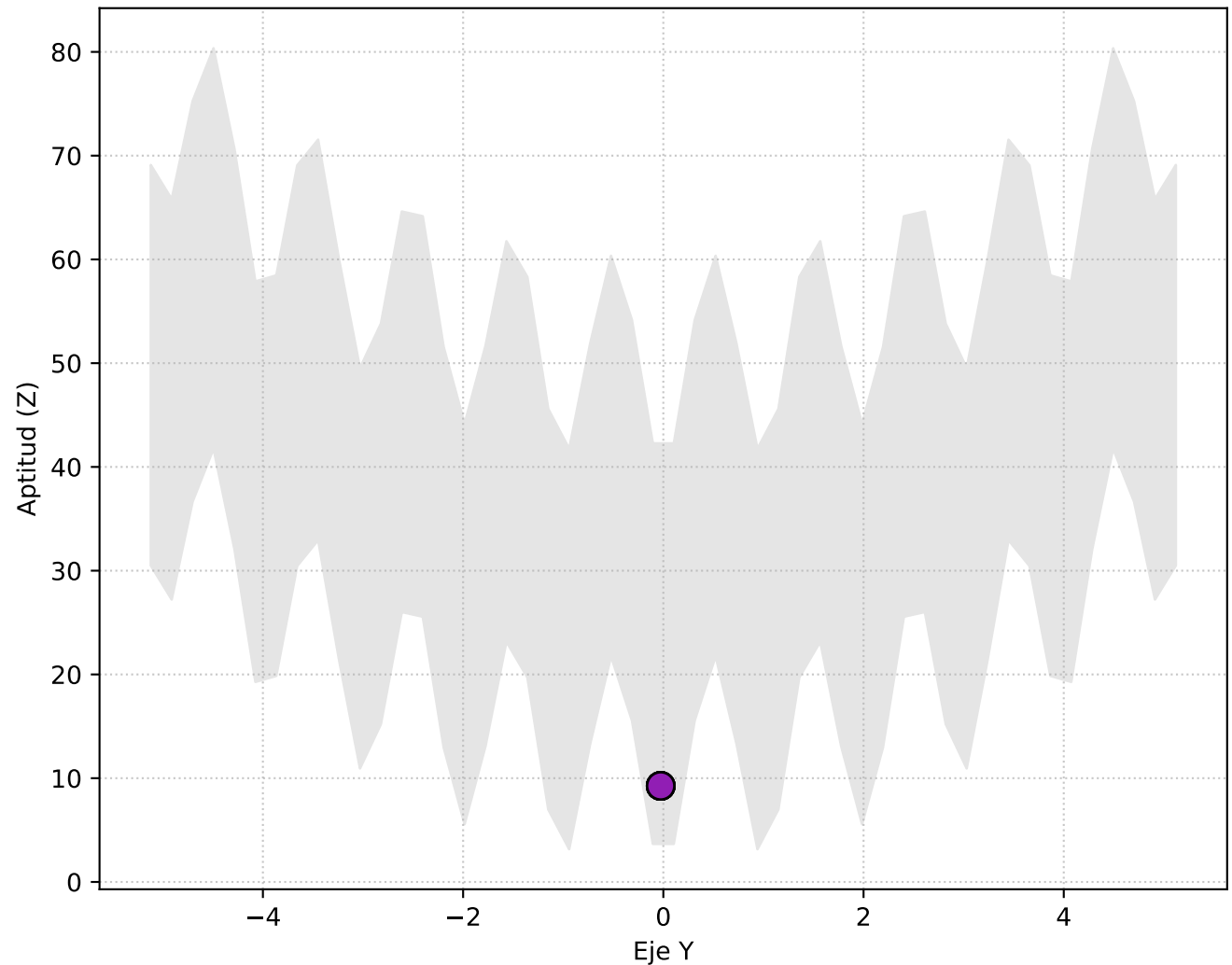
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

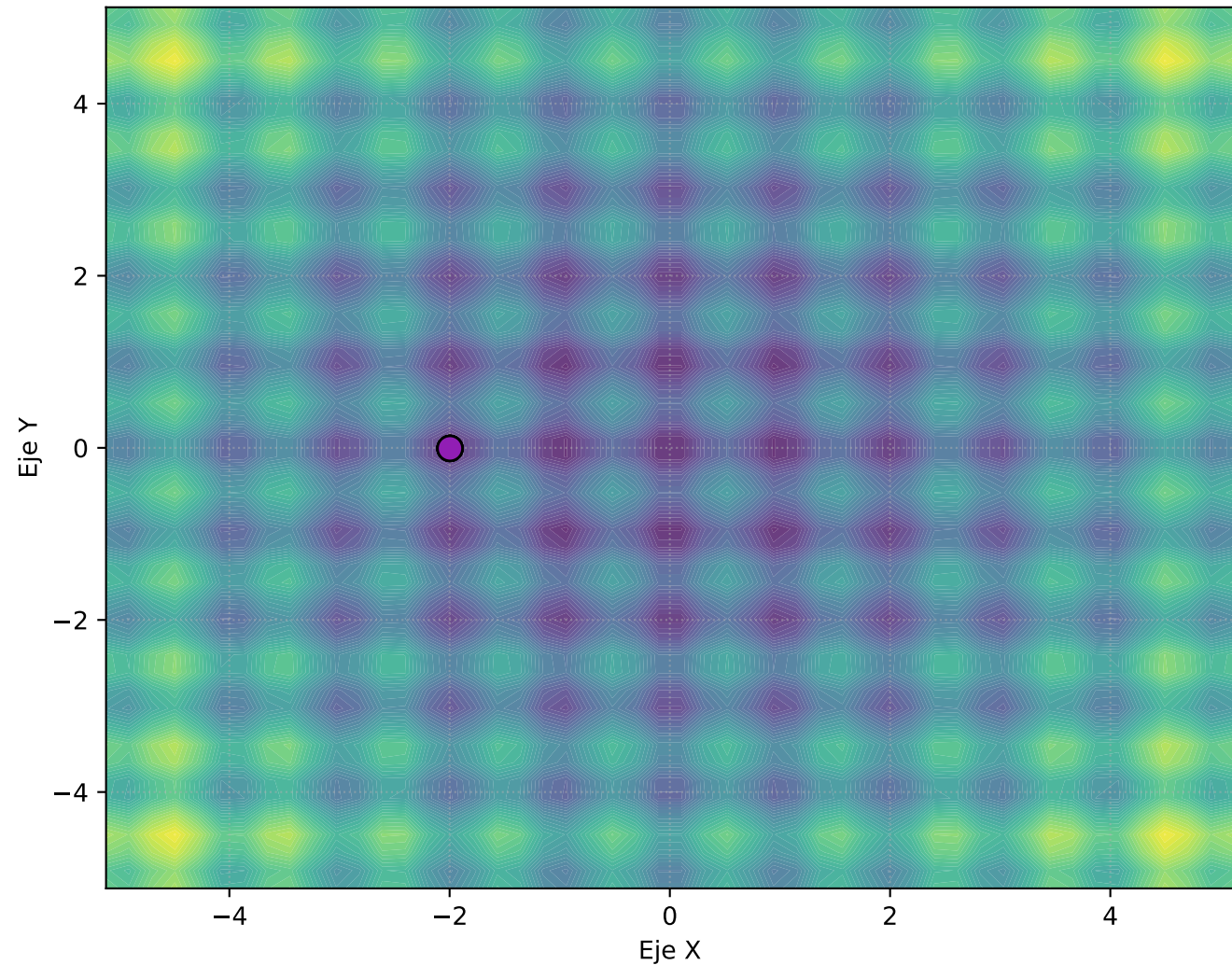


Vista Frontal (Y vs Elevación Z)

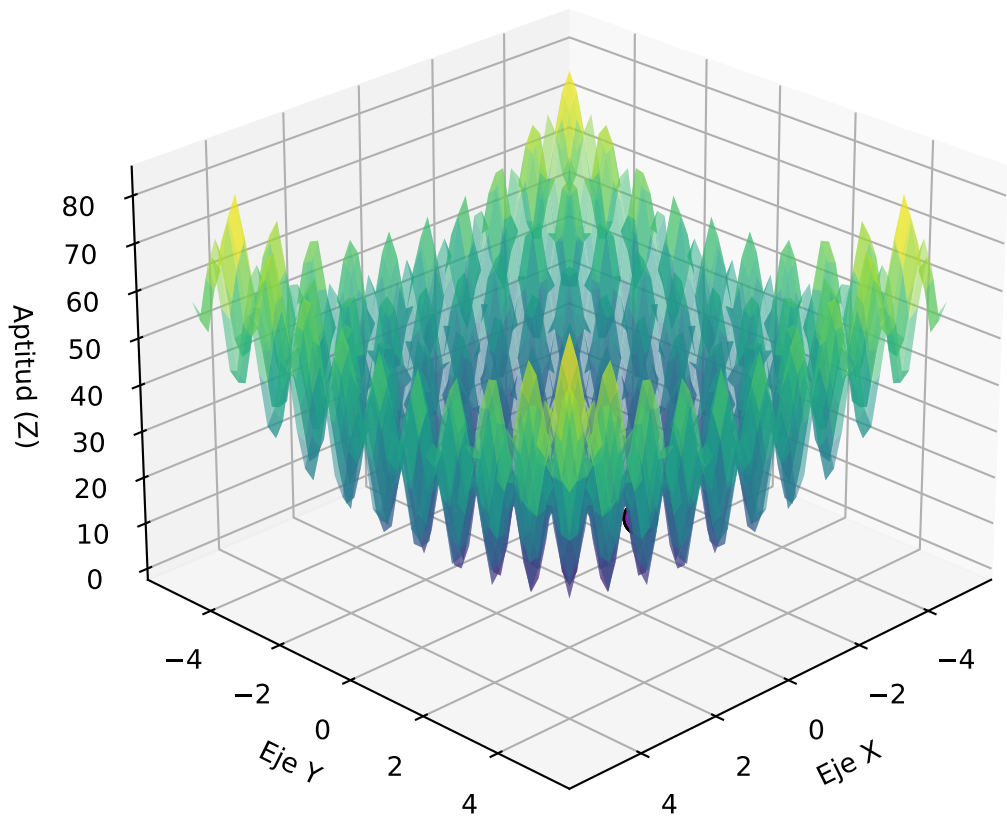


Progreso: Generación 30

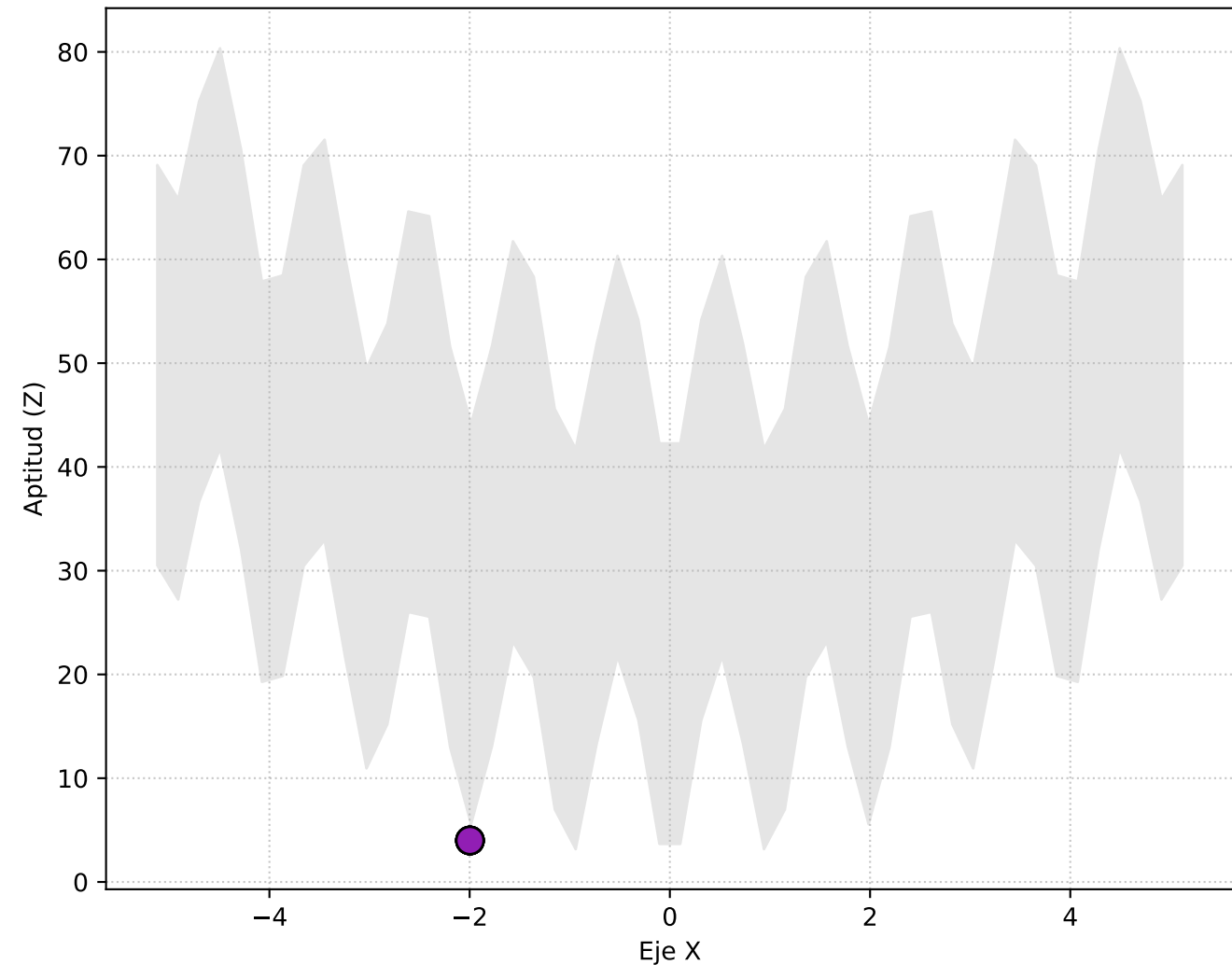
Vista Superior (X, Y)



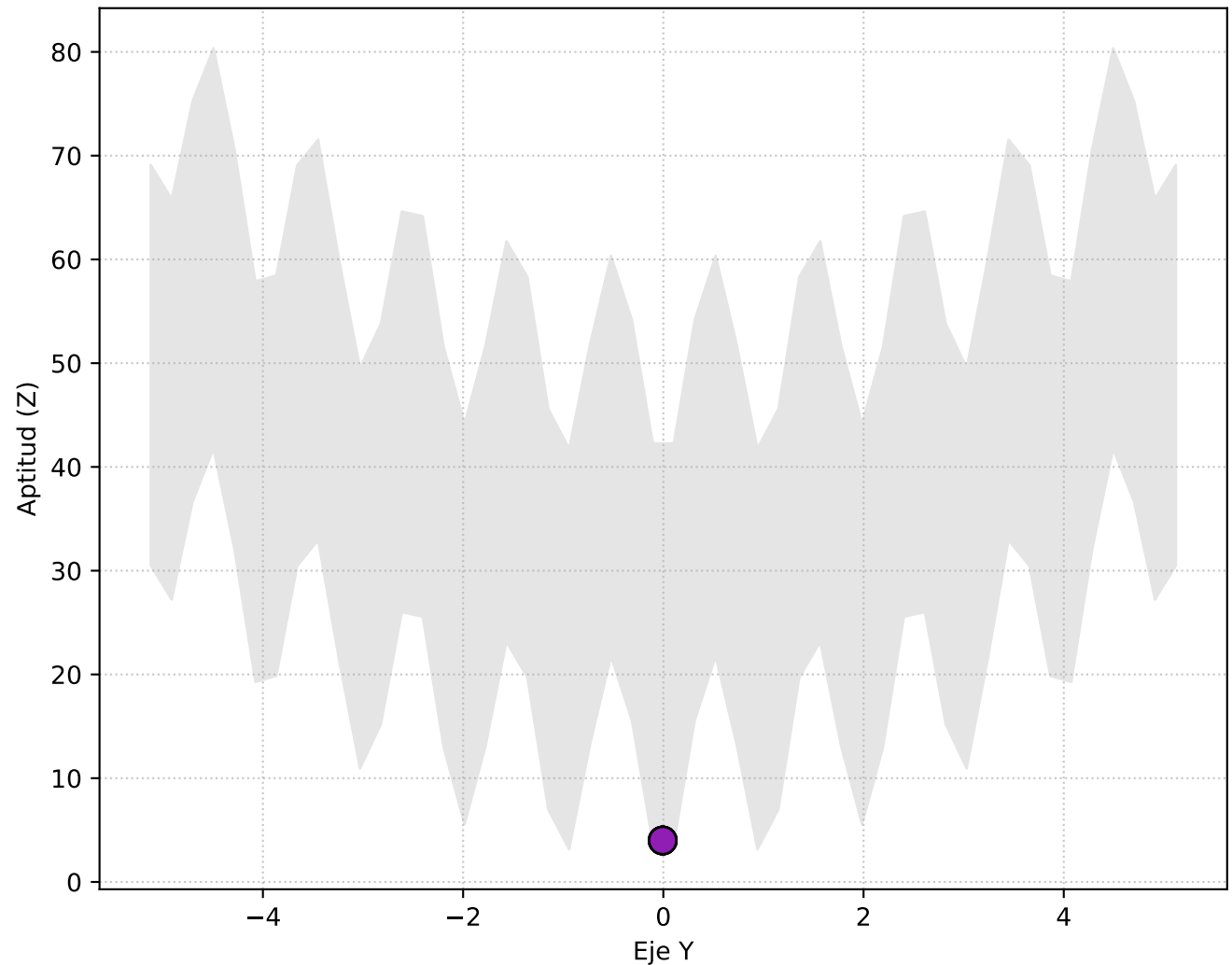
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

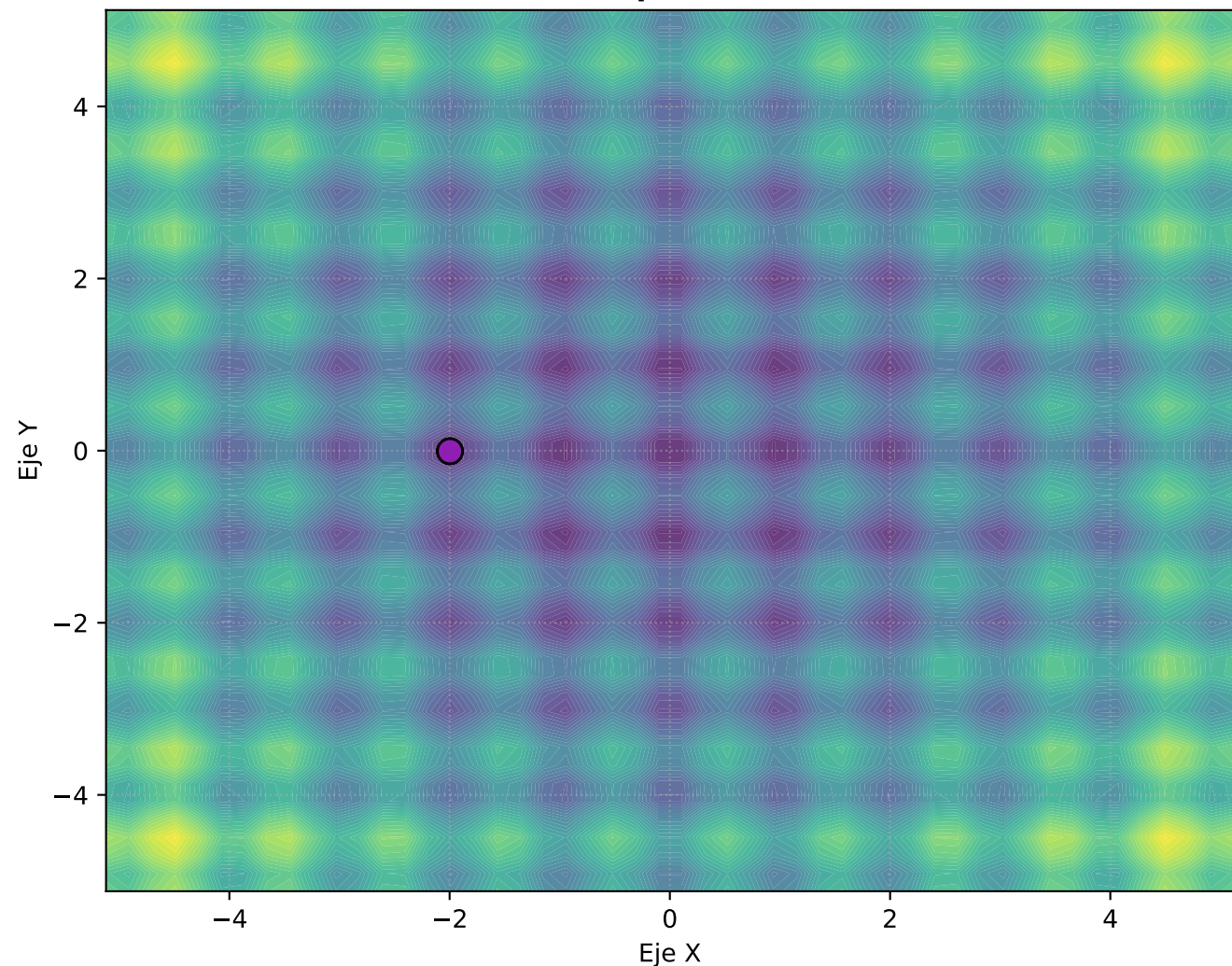


Vista Frontal (Y vs Elevación Z)

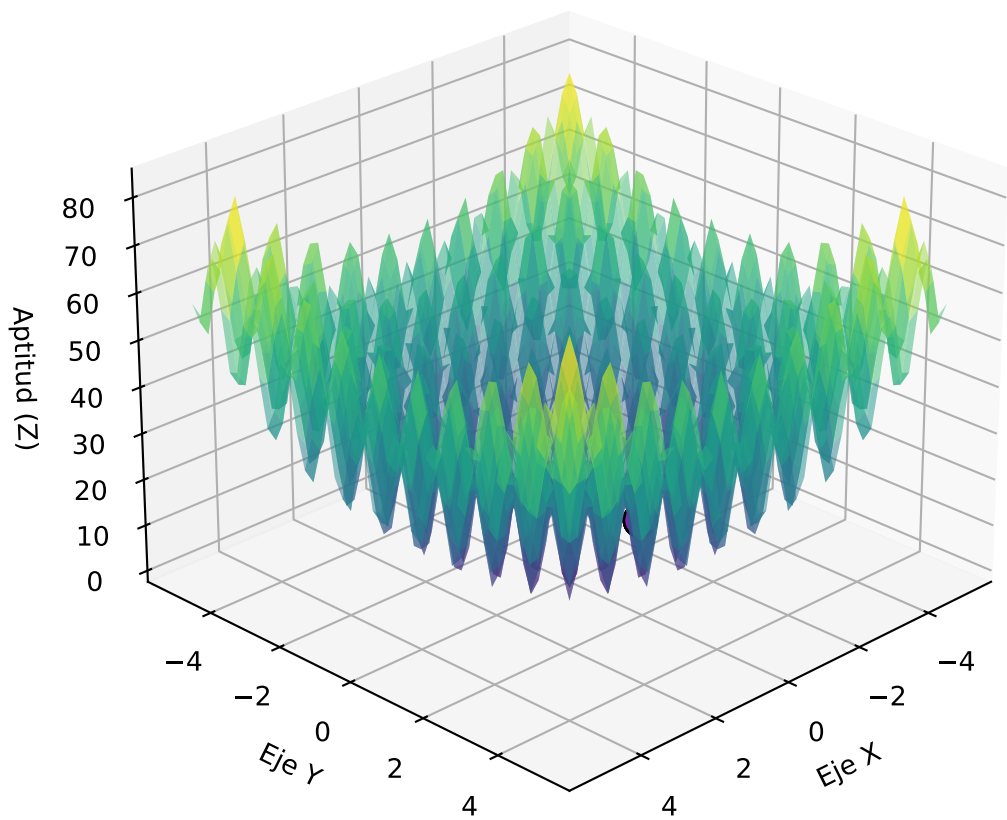


Progreso: Generación 40

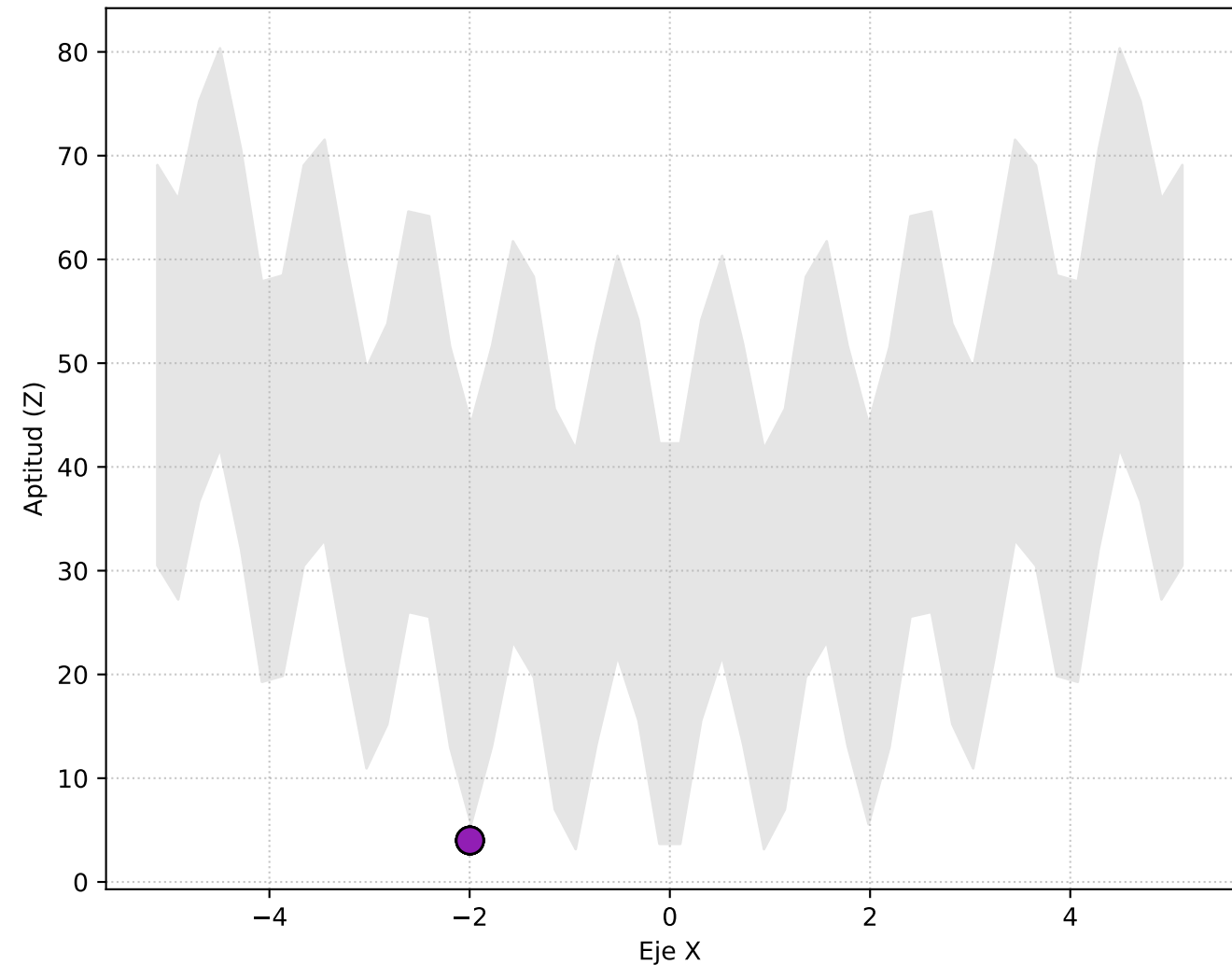
Vista Superior (X, Y)



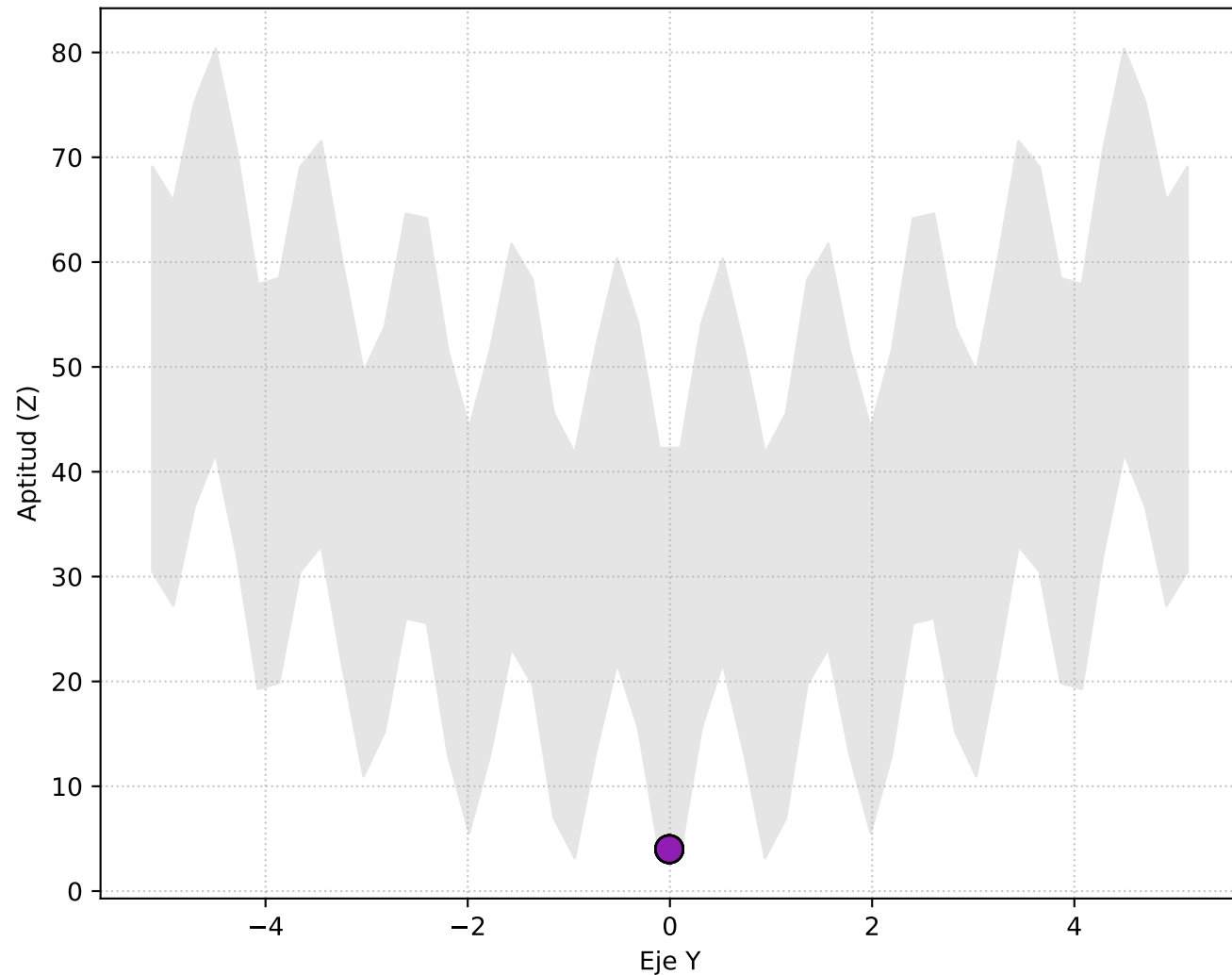
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

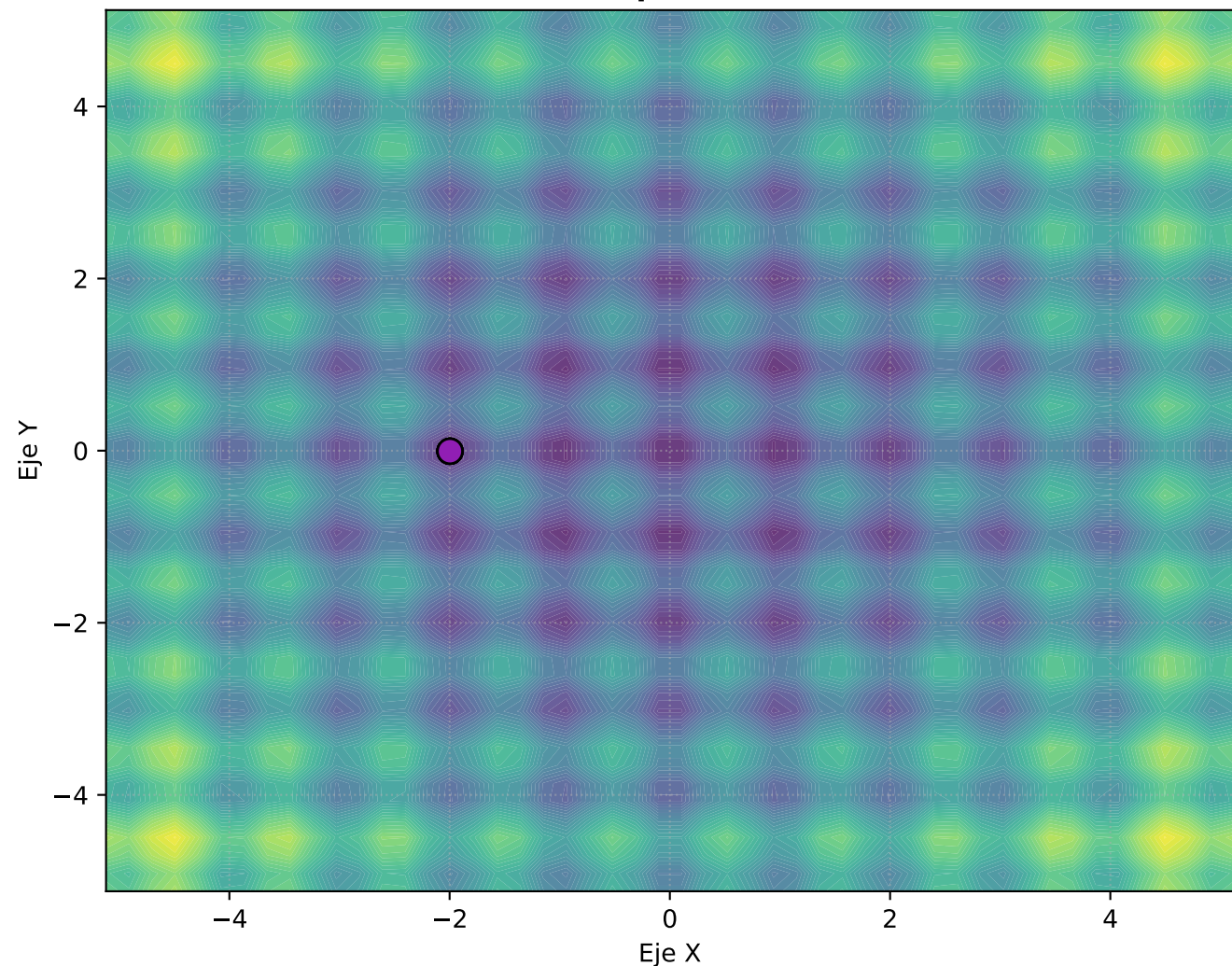


Vista Frontal (Y vs Elevación Z)

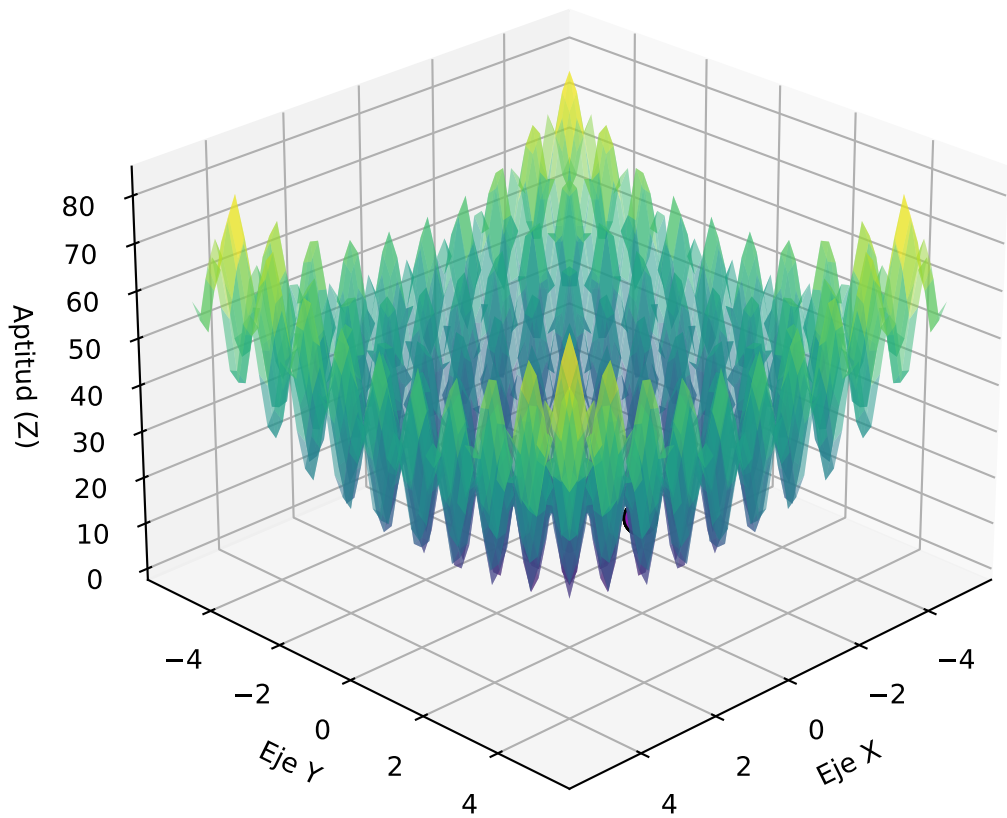


Progreso: Generación 50

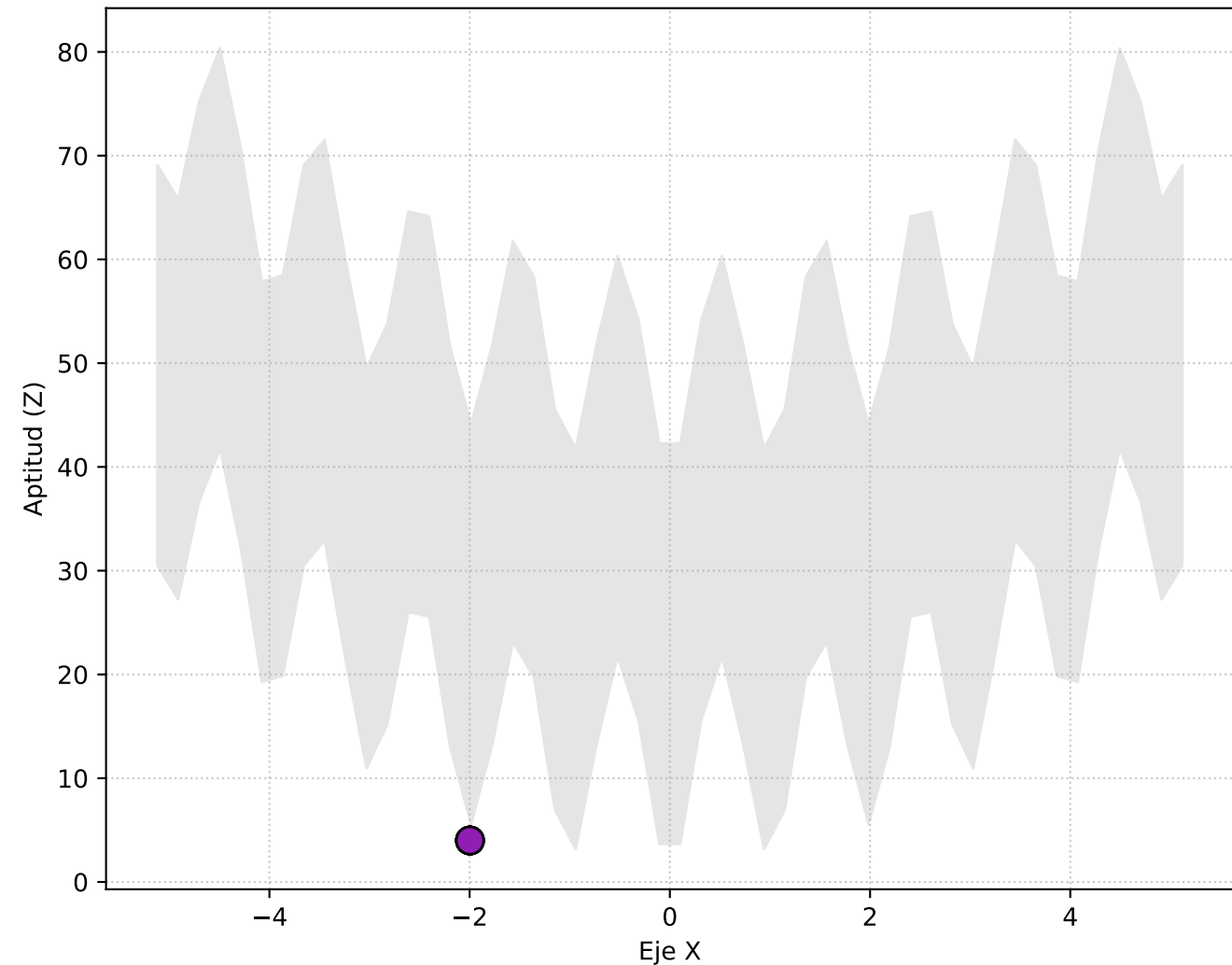
Vista Superior (X, Y)



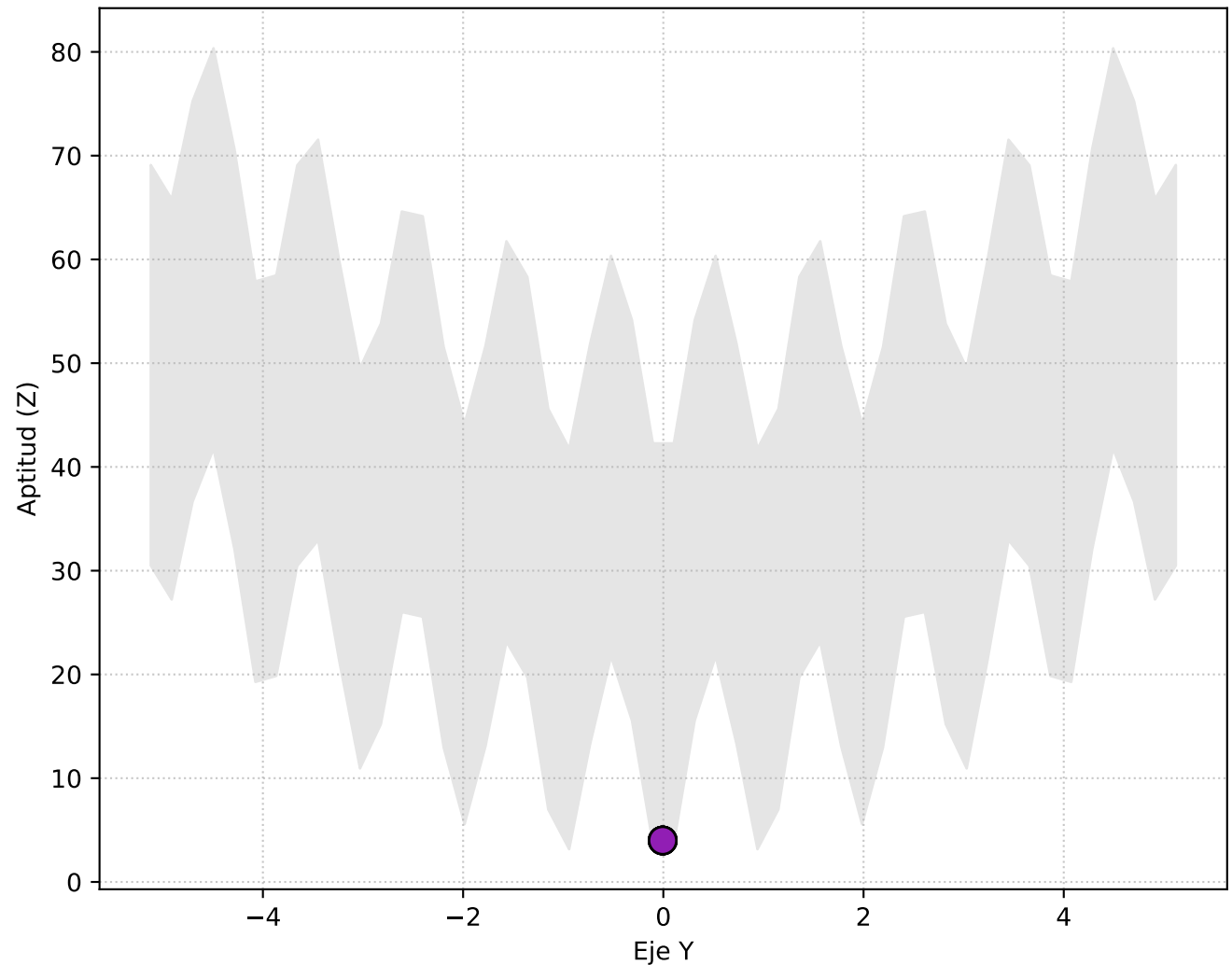
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

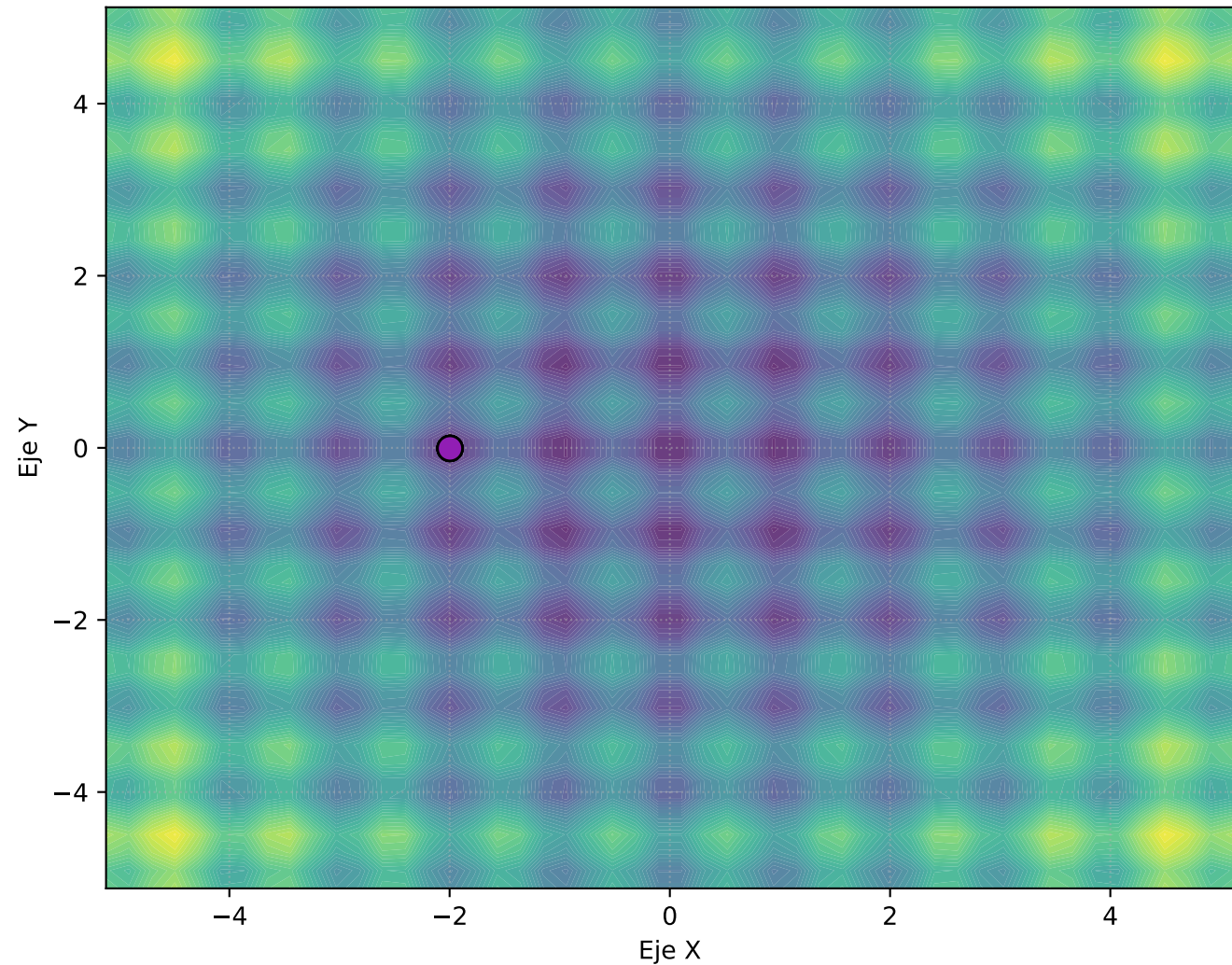


Vista Frontal (Y vs Elevación Z)

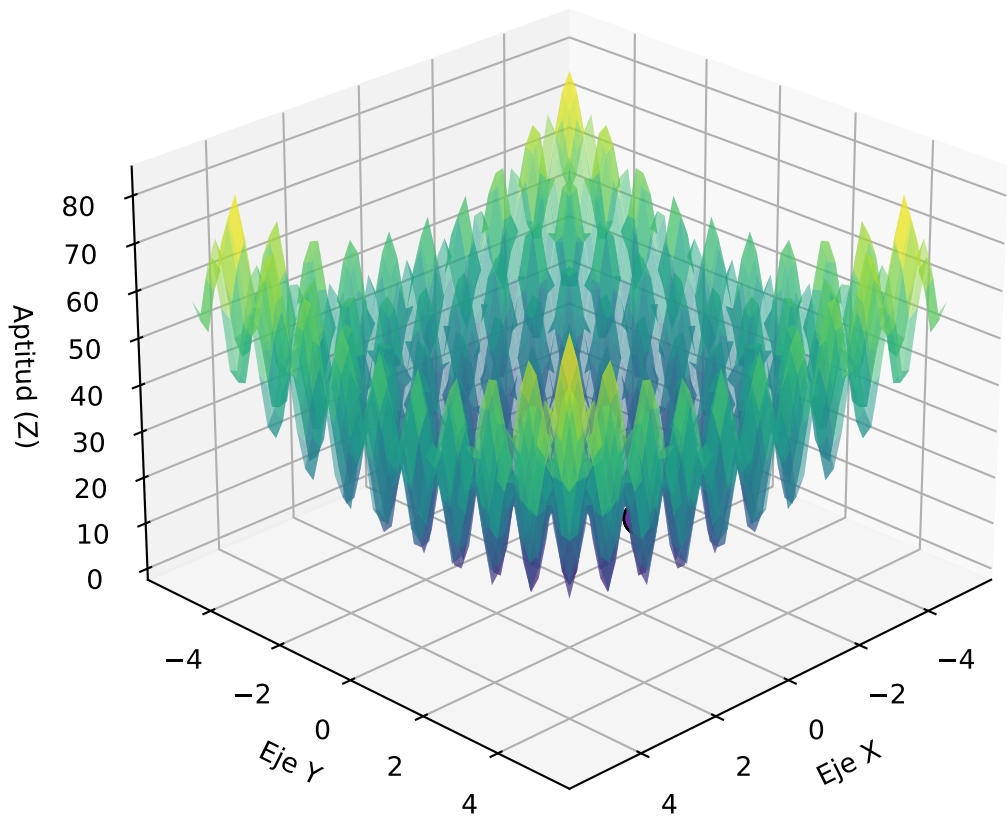


Progreso: Generación 60

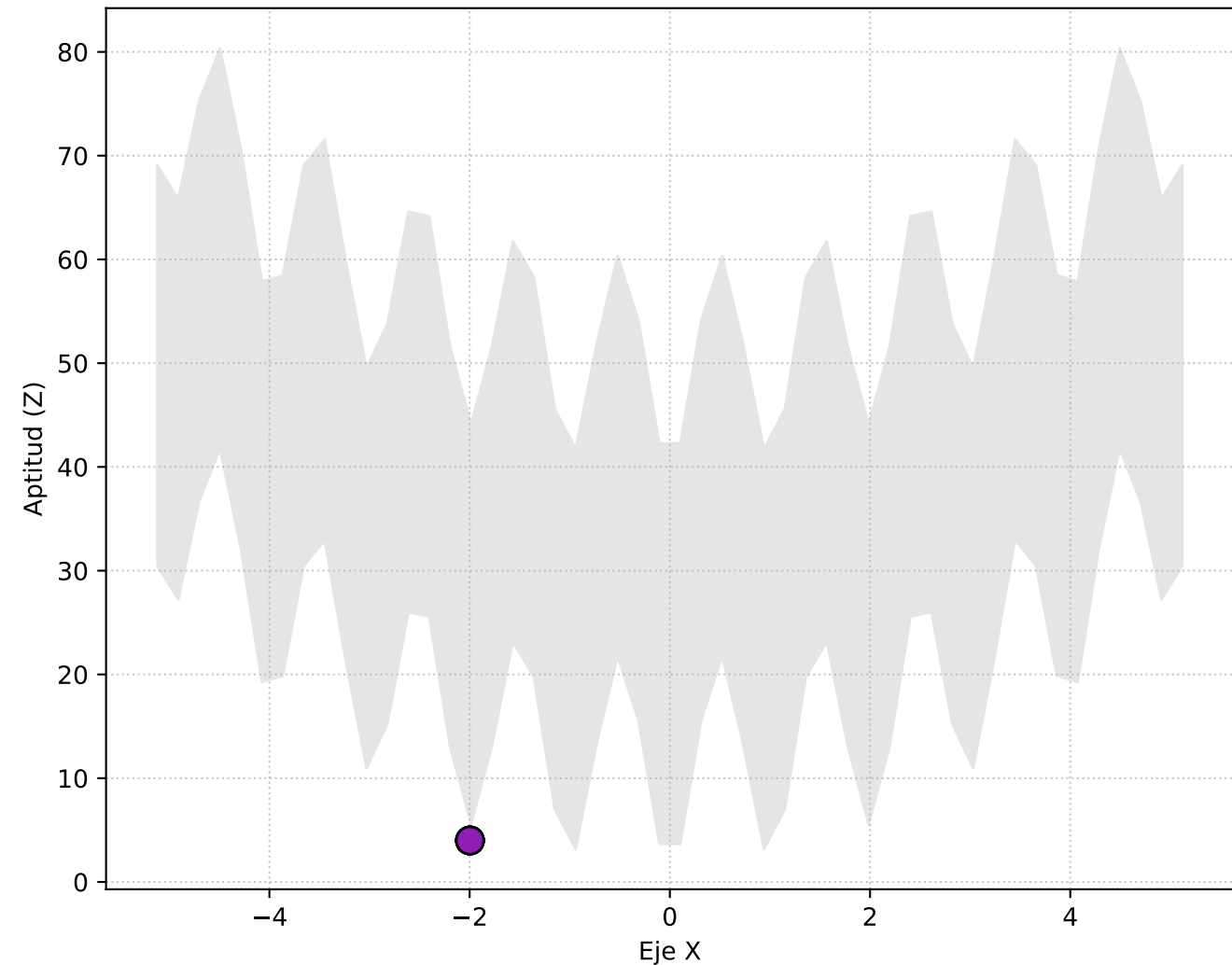
Vista Superior (X, Y)



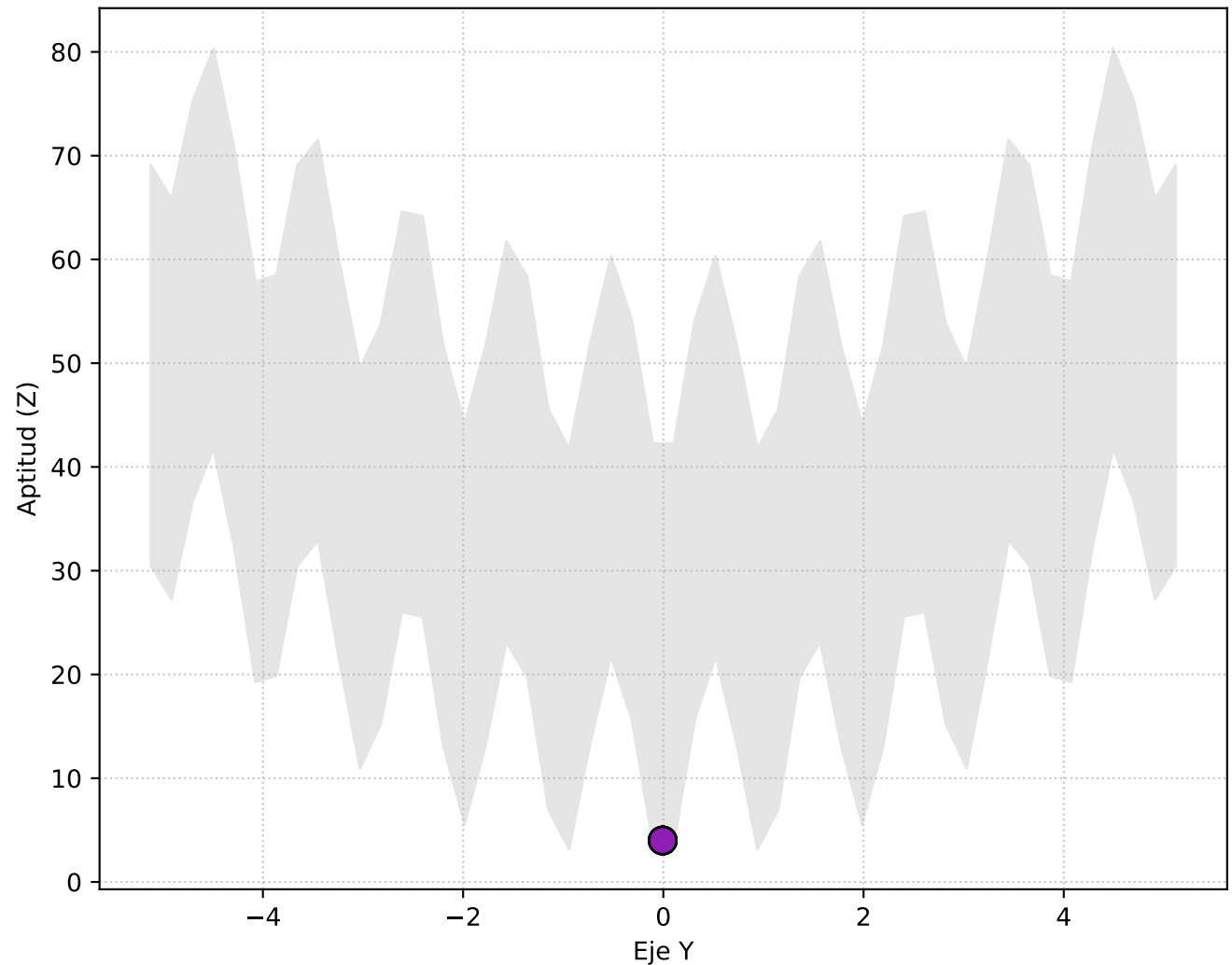
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

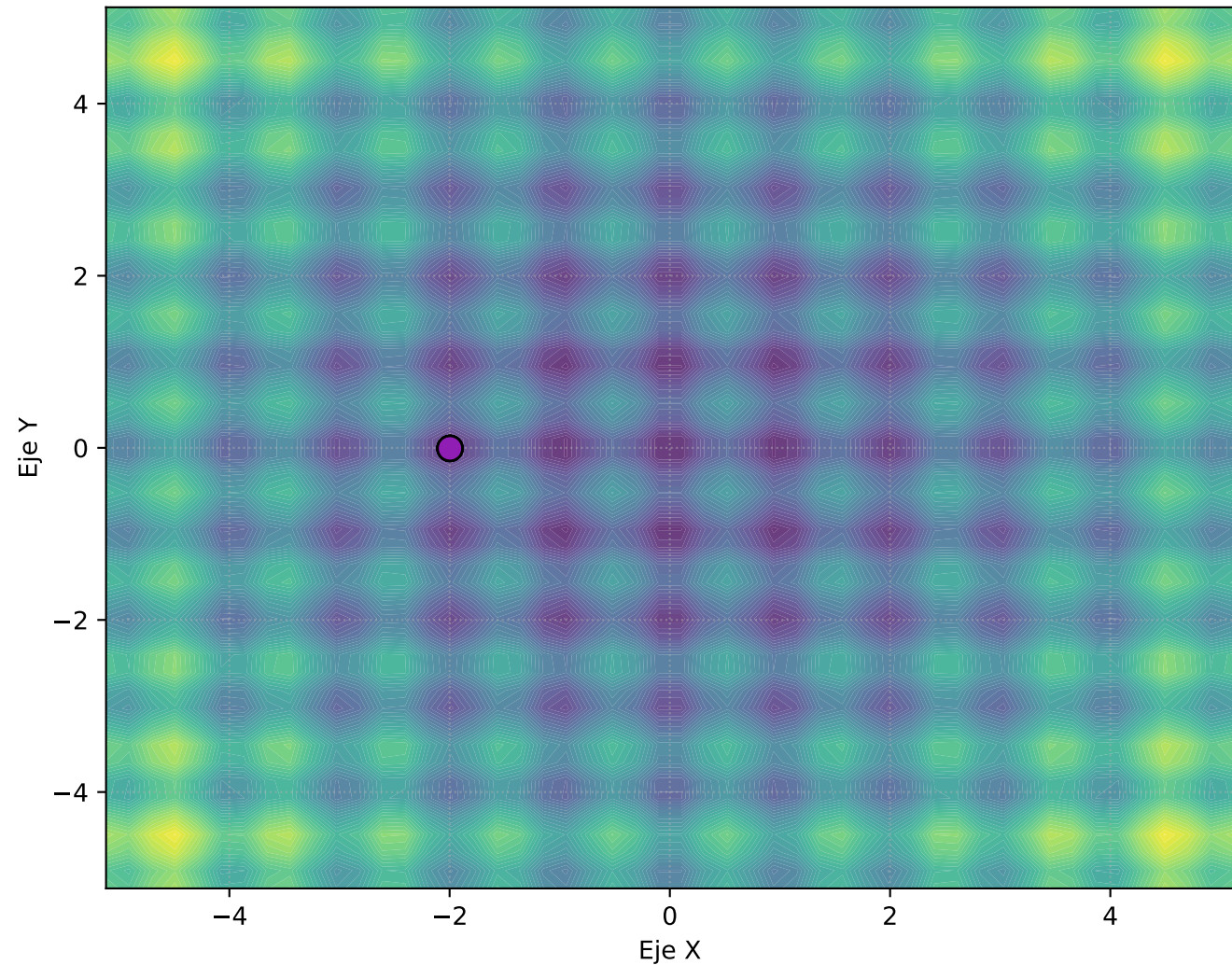


Vista Frontal (Y vs Elevación Z)

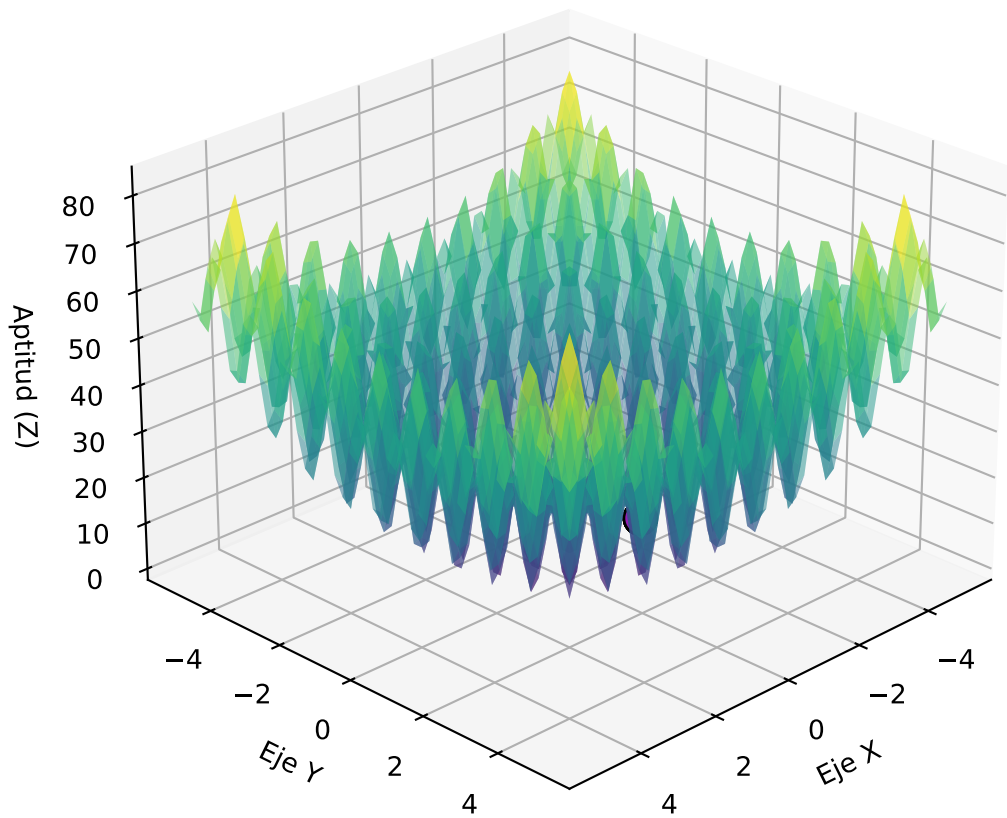


Progreso: Generación 70

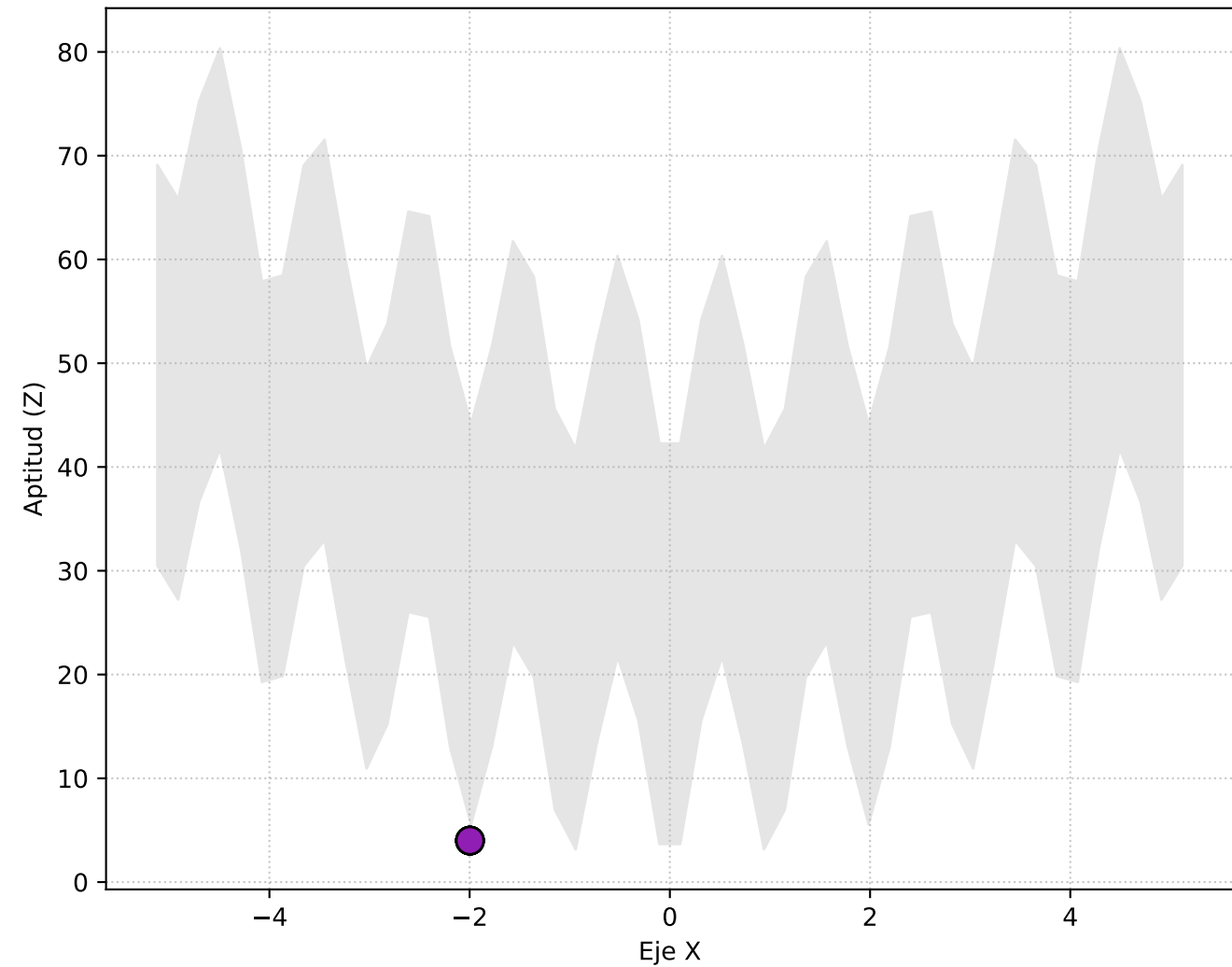
Vista Superior (X, Y)



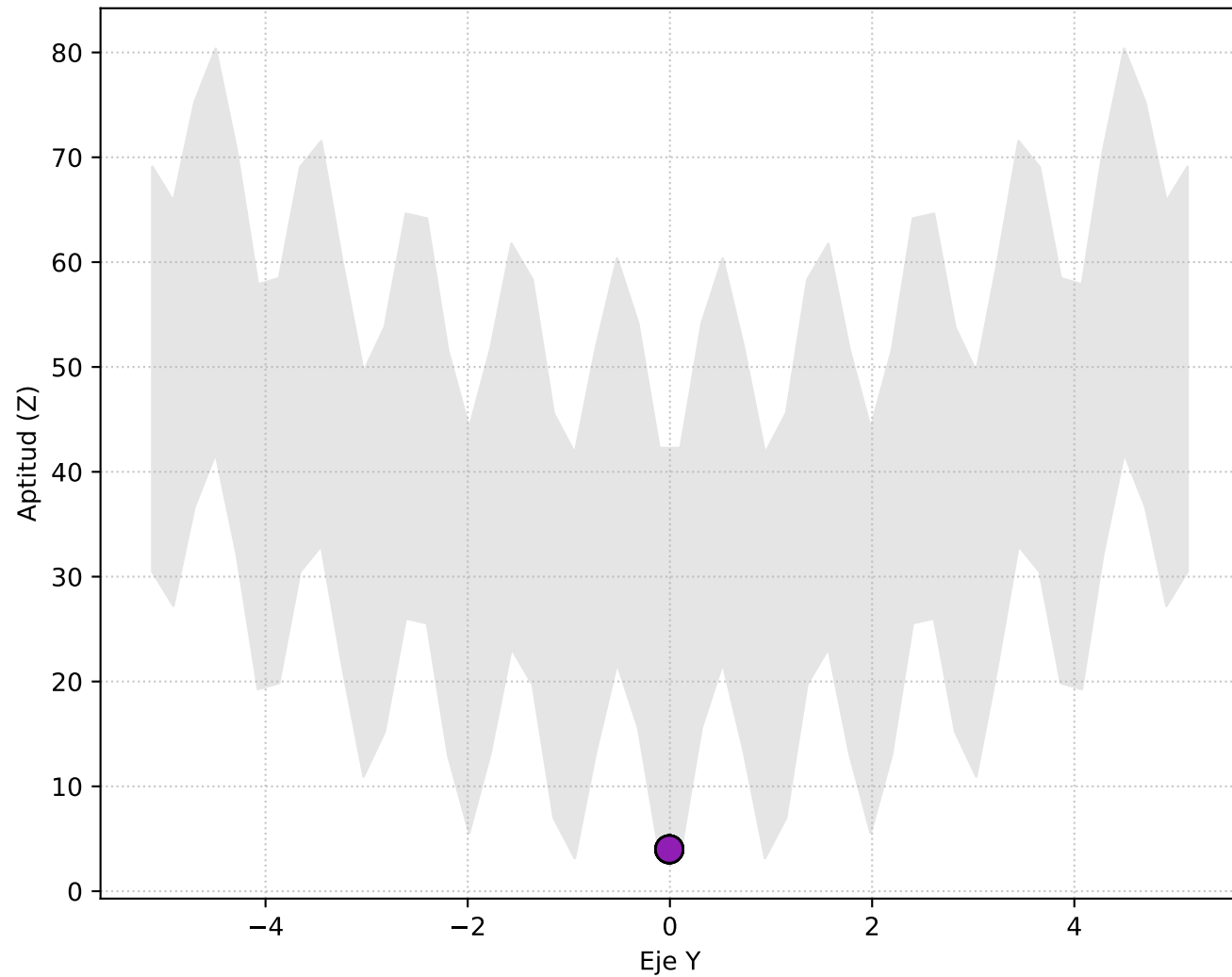
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

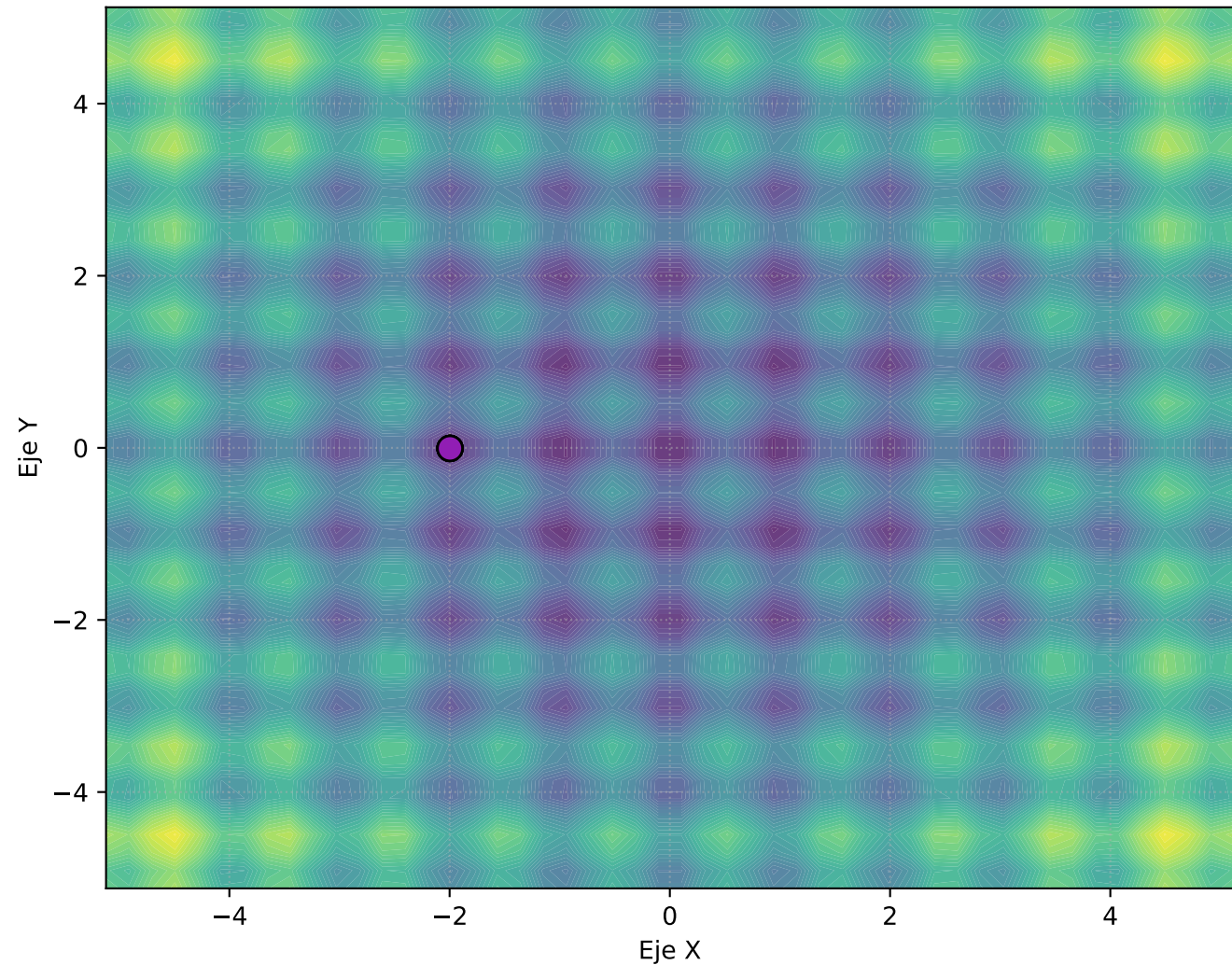


Vista Frontal (Y vs Elevación Z)

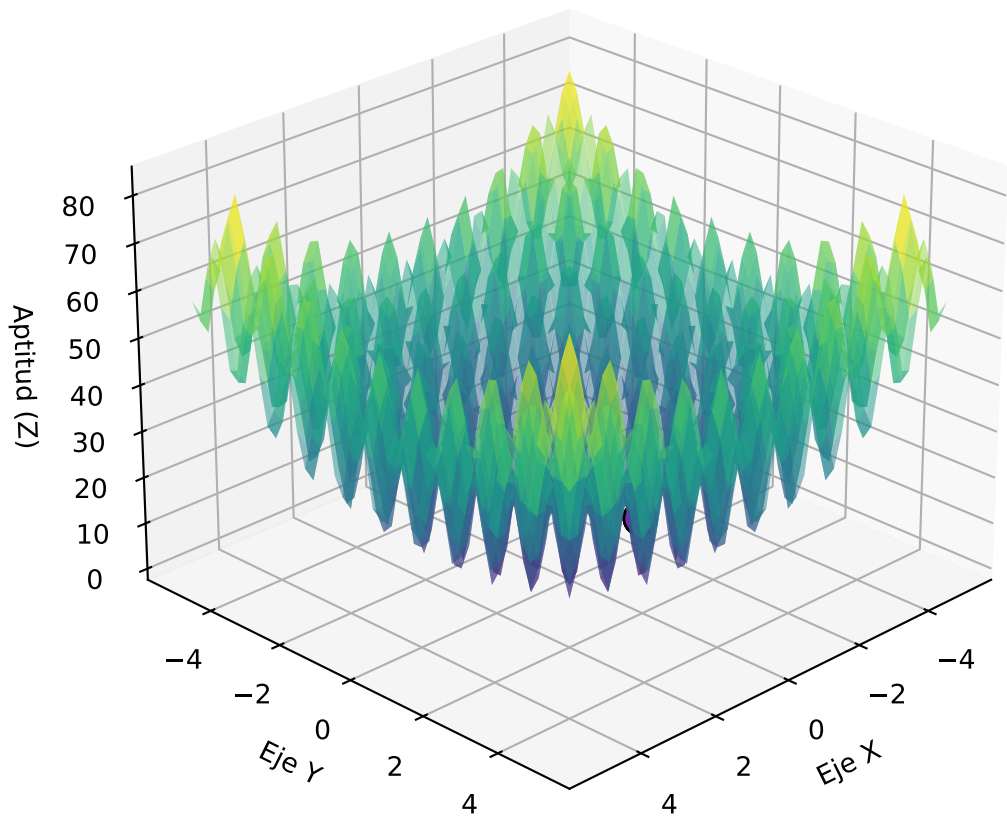


Progreso: Generación 80

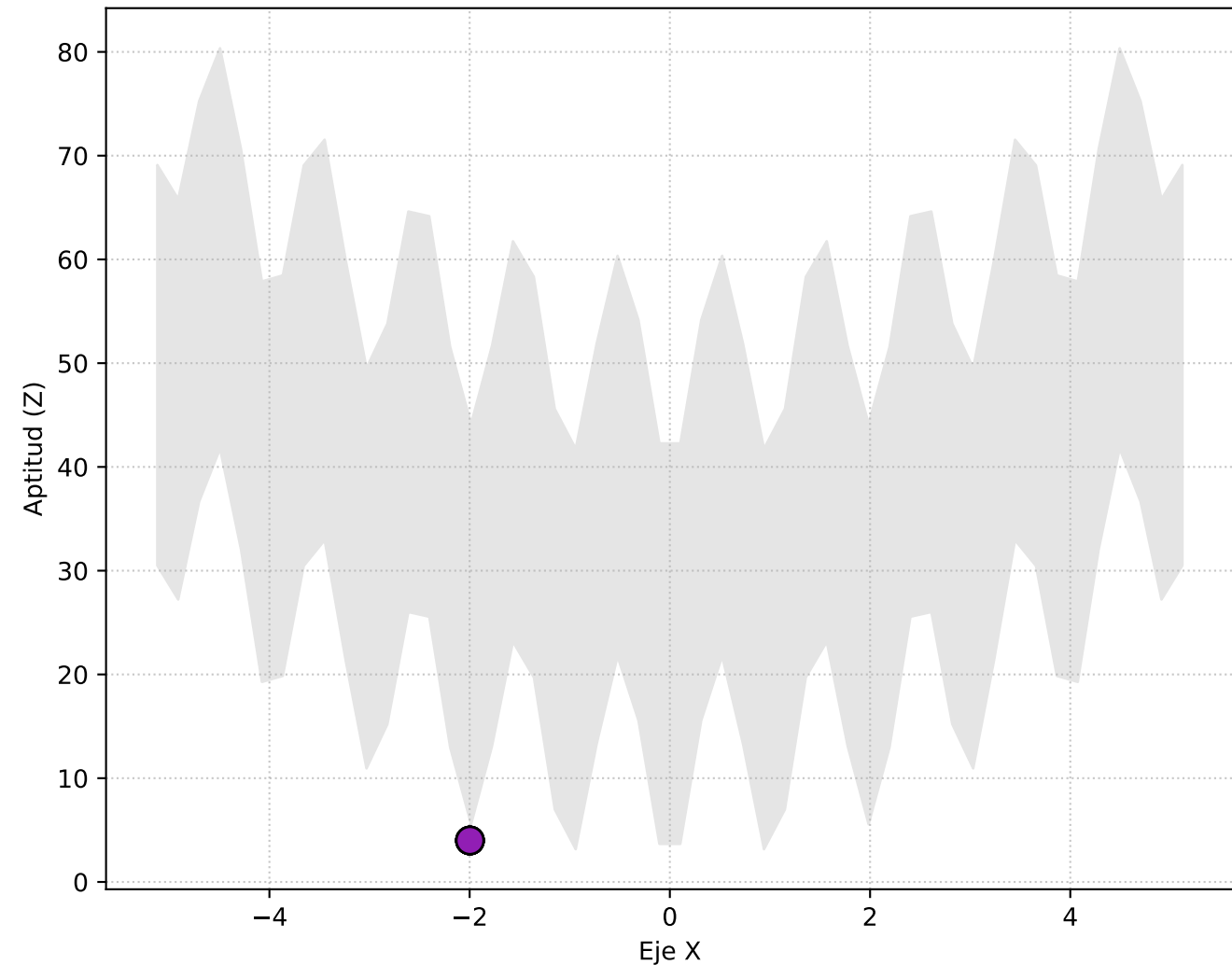
Vista Superior (X, Y)



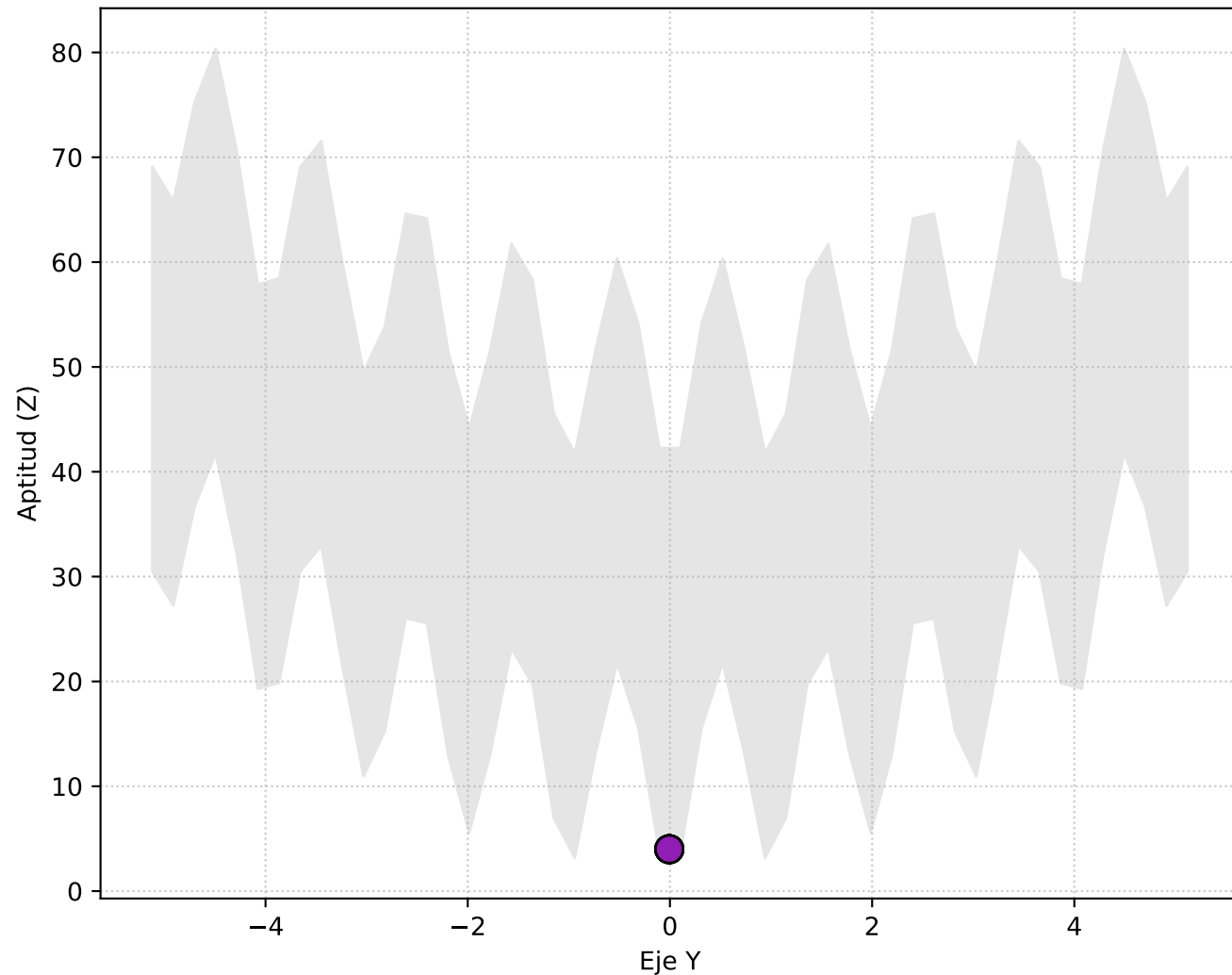
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

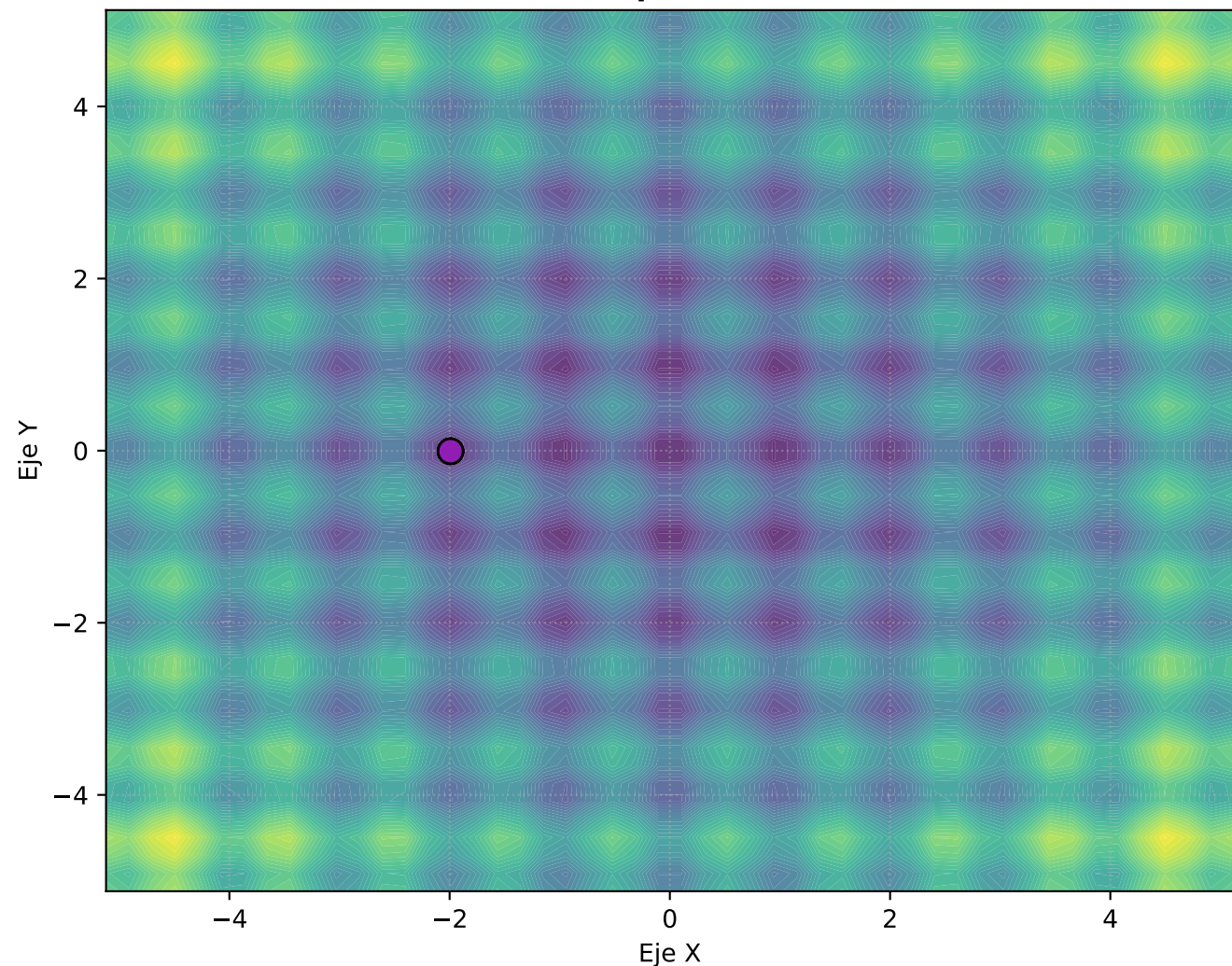


Vista Frontal (Y vs Elevación Z)

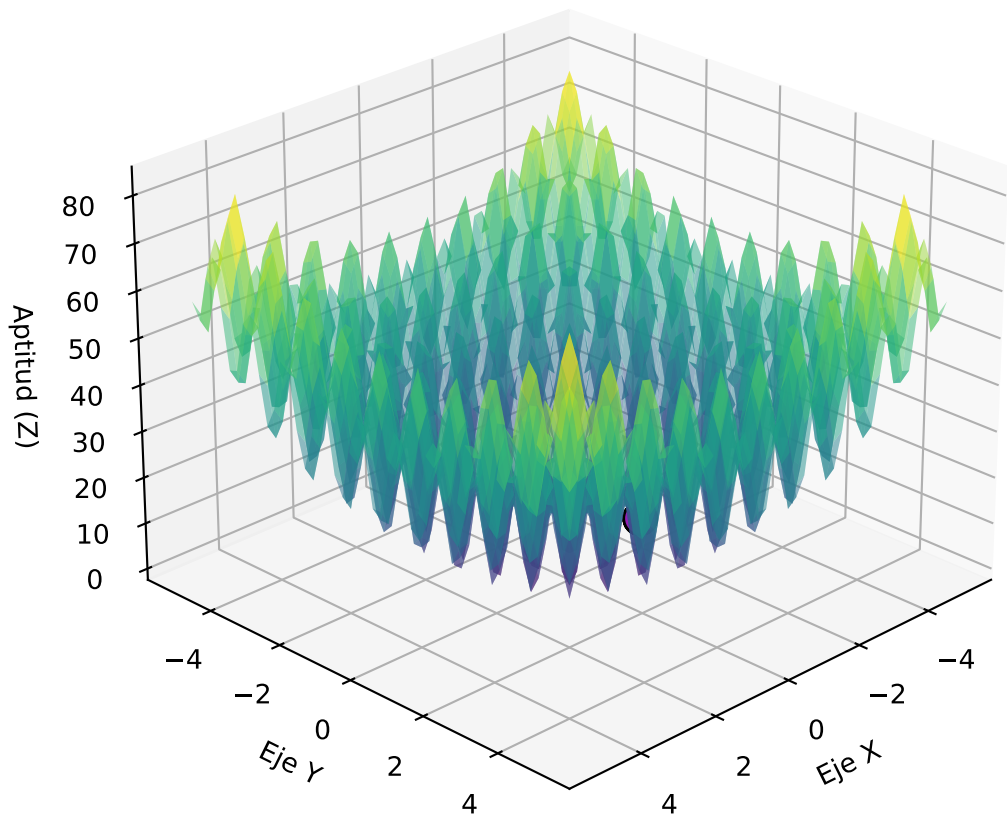


Progreso: Generación 90

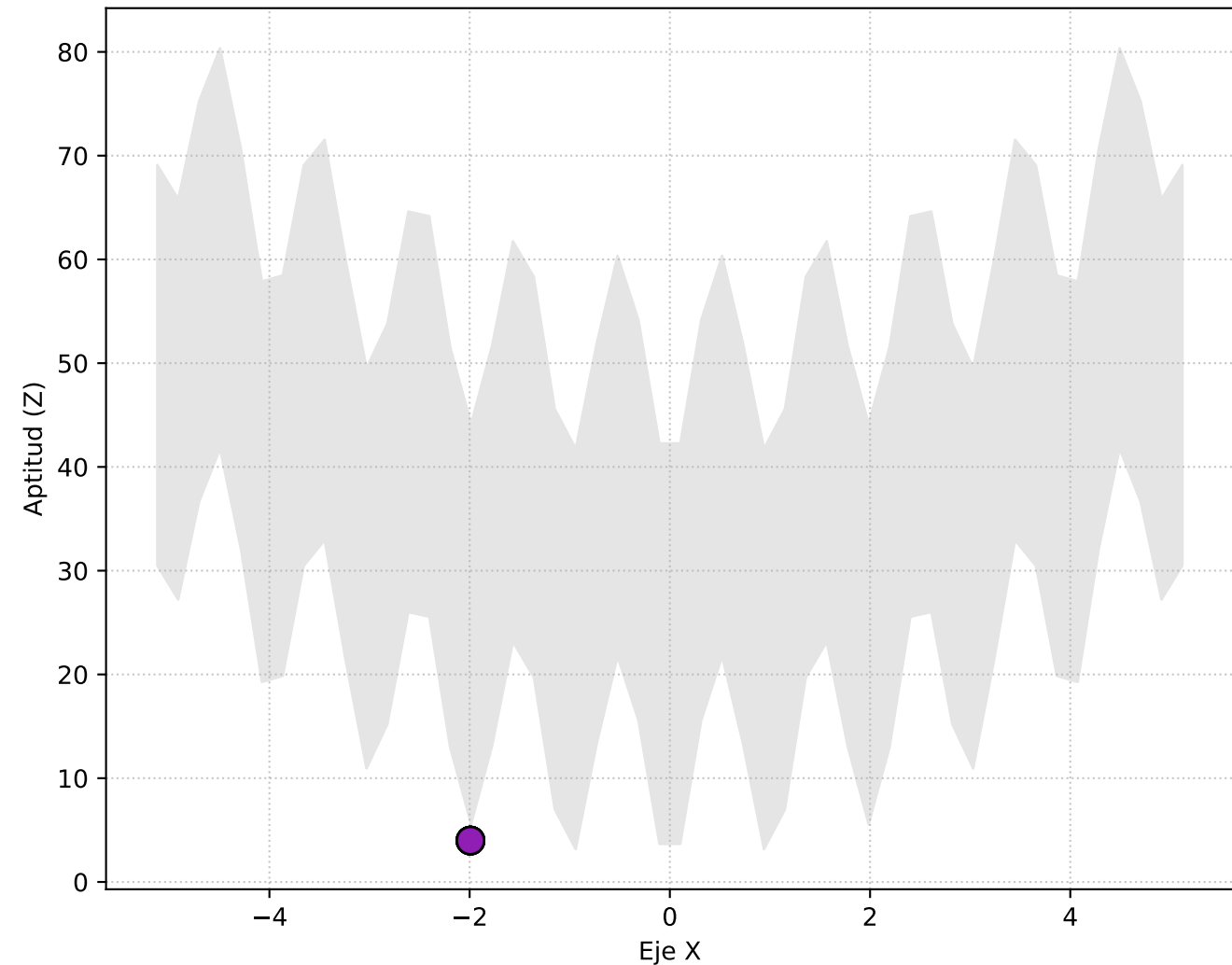
Vista Superior (X, Y)



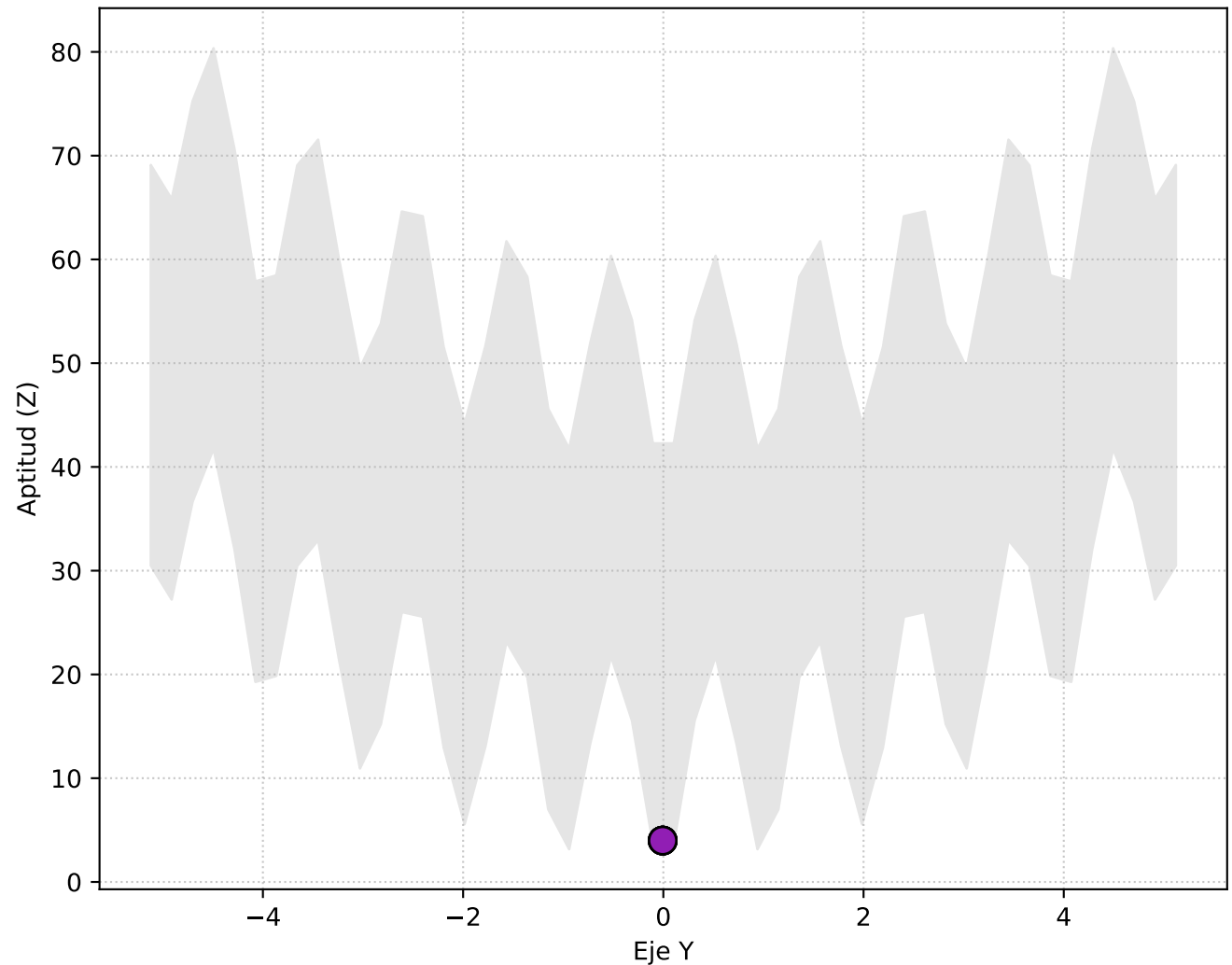
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

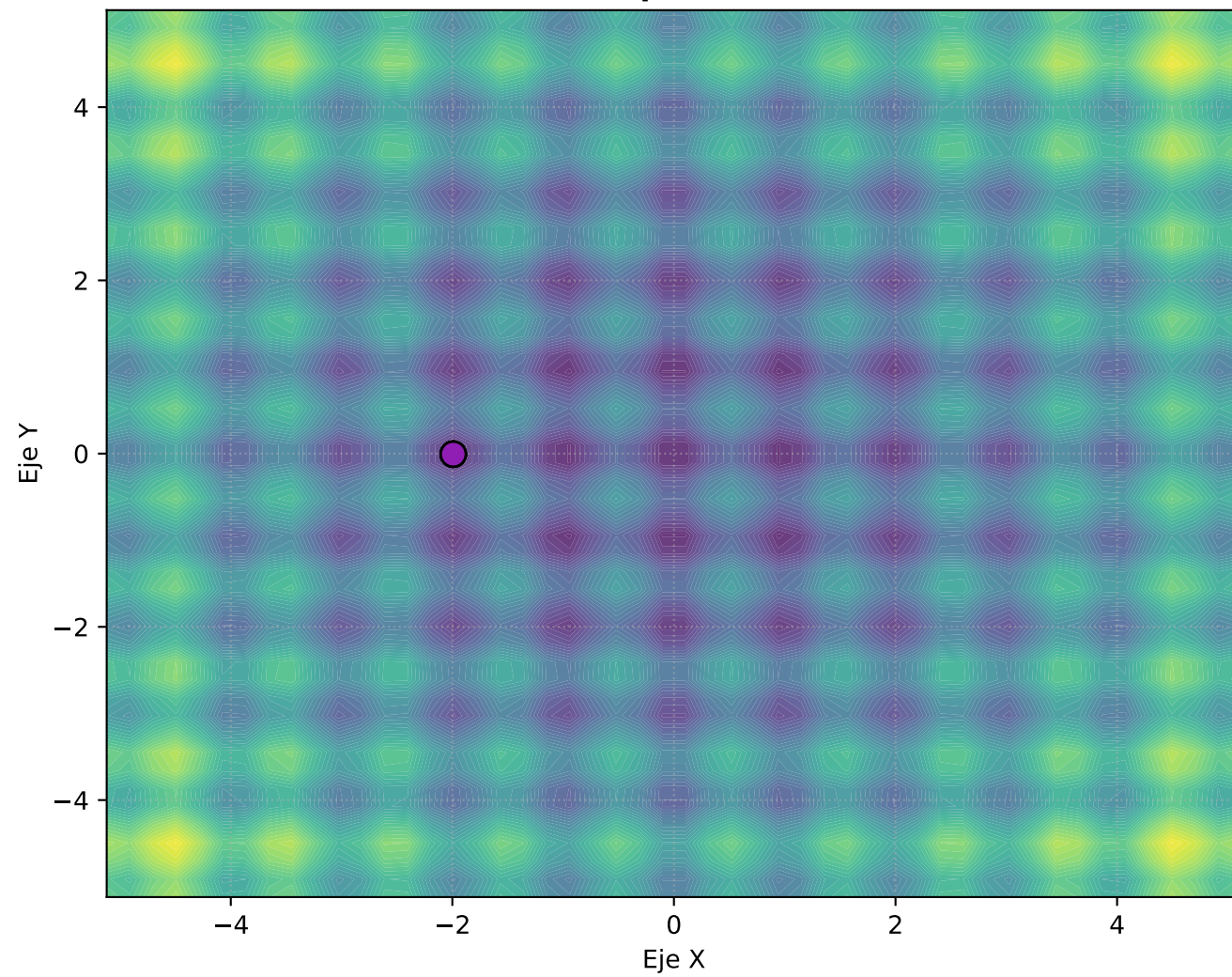


Vista Frontal (Y vs Elevación Z)

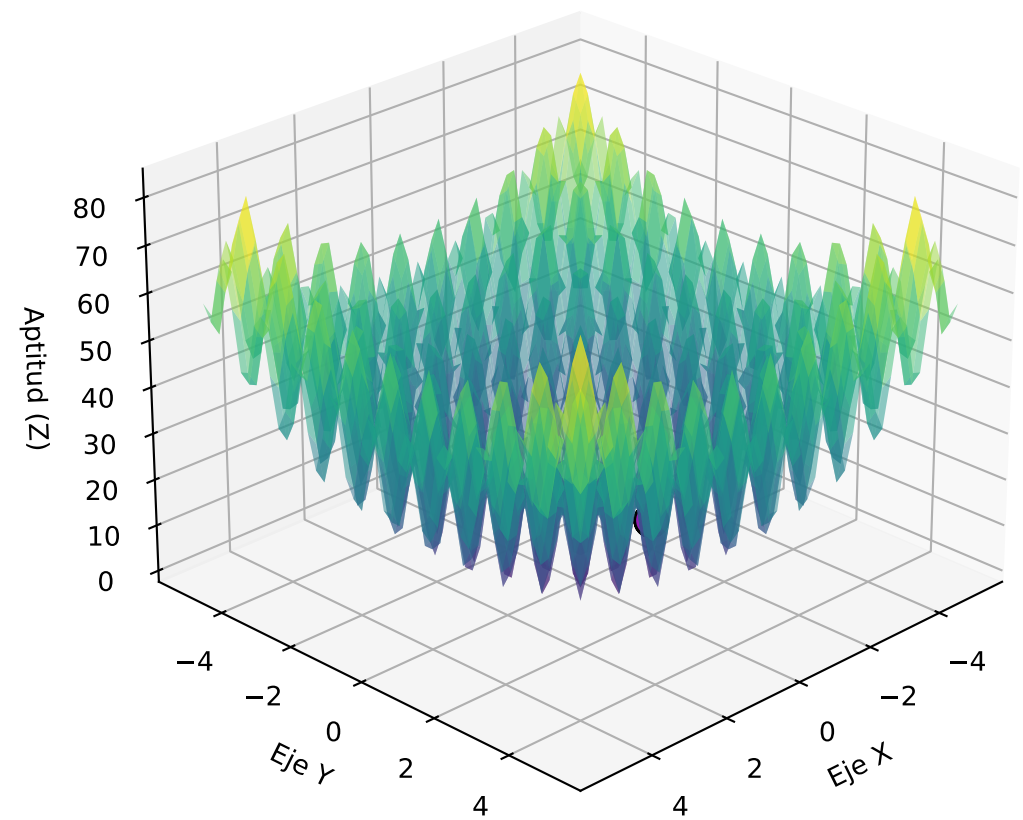


Progreso: Generación 100

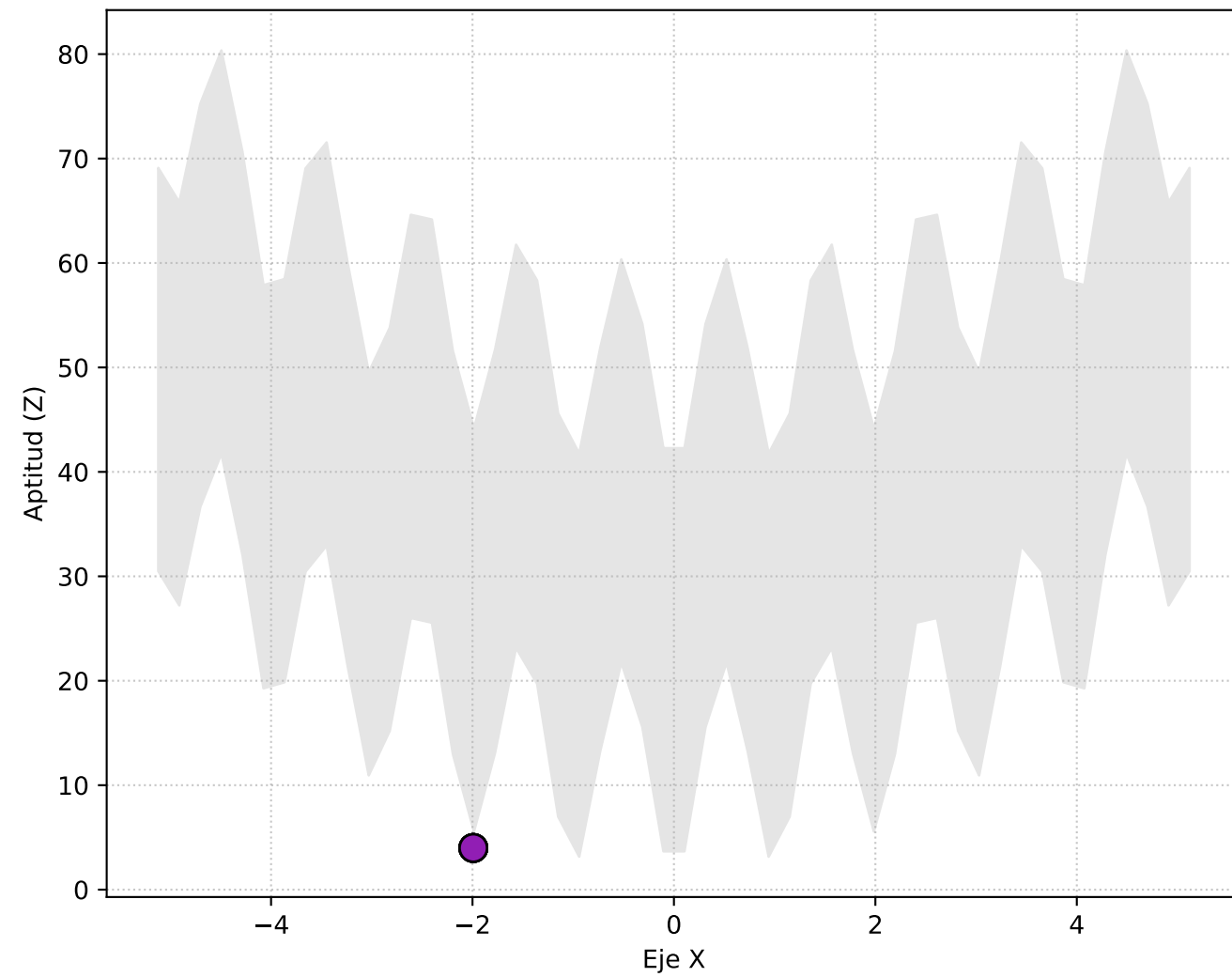
Vista Superior (X, Y)



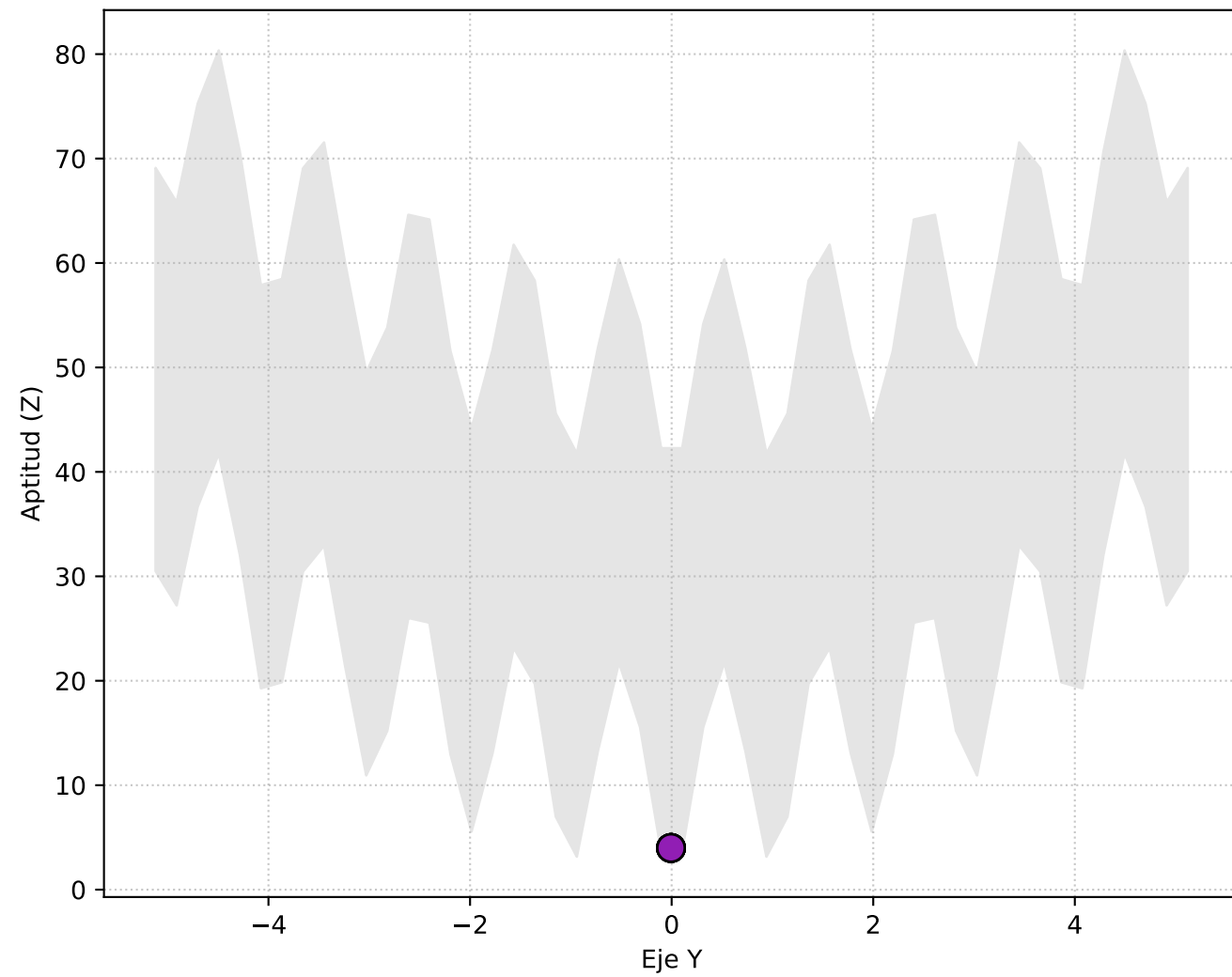
Perspectiva 3D



Vista Lateral (X vs Elevación Z)



Vista Frontal (Y vs Elevación Z)

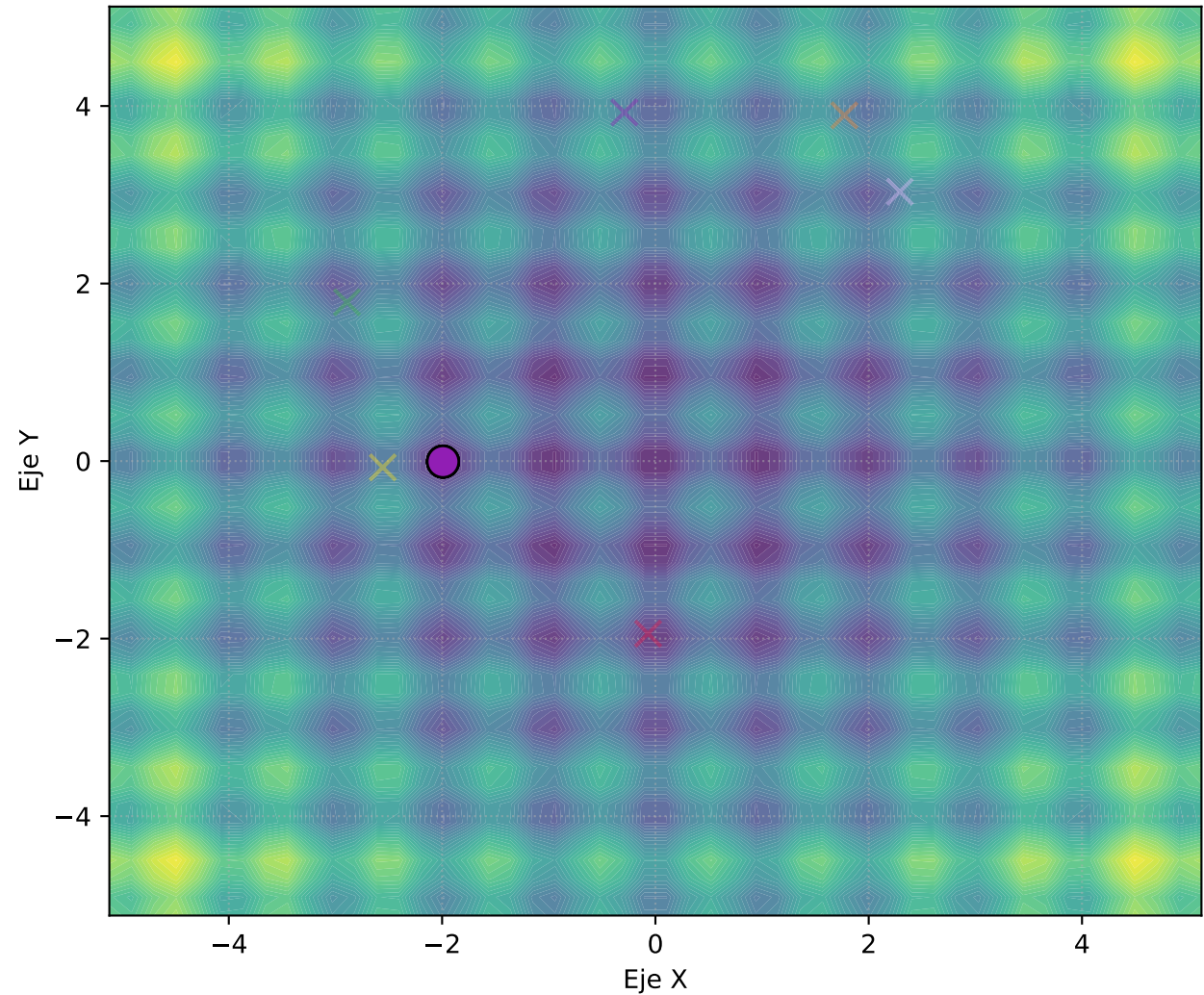


Posición del Mejor Individuo por Salto

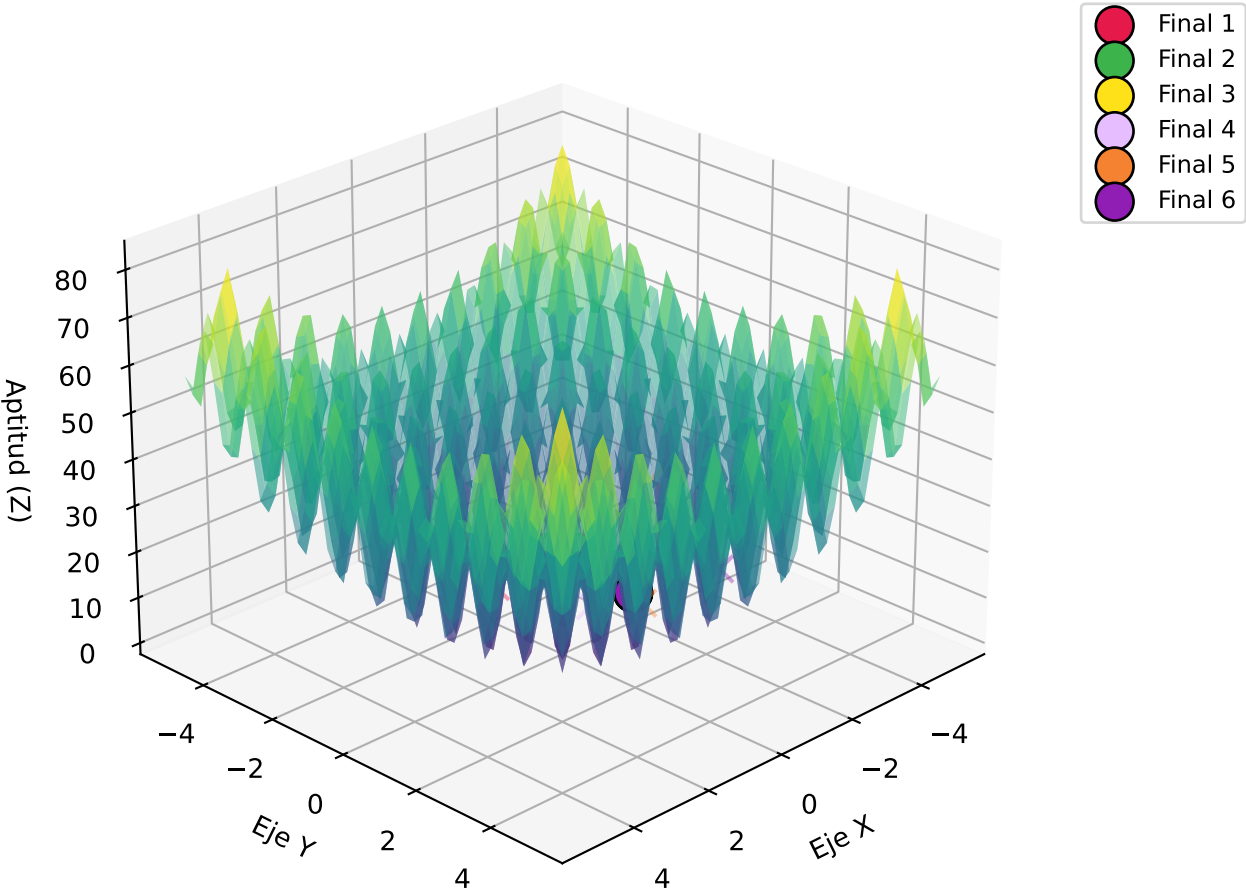
Gen	Mejores Bits	Coordenadas (X, Y)	Aptitud
10	[0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1]	X: -2.46, Y: -0.06	0.0365
20	[0, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1]	X: -2.16, Y: -0.03	0.0975
30	[0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]	X: -2.00, Y: -0.01	0.2001
40	[0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]	X: -2.00, Y: -0.01	0.2001
50	[0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]	X: -2.00, Y: -0.01	0.2001
60	[0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]	X: -2.00, Y: -0.01	0.2001
70	[0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]	X: -2.00, Y: -0.01	0.2001
80	[0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]	X: -2.00, Y: -0.01	0.2001
90	[0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]	X: -1.99, Y: -0.01	0.2005
100	[0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]	X: -1.99, Y: -0.01	0.2005

Comparativa: Iniciales (Tache) vs Finales Gen 100 (Círculo)

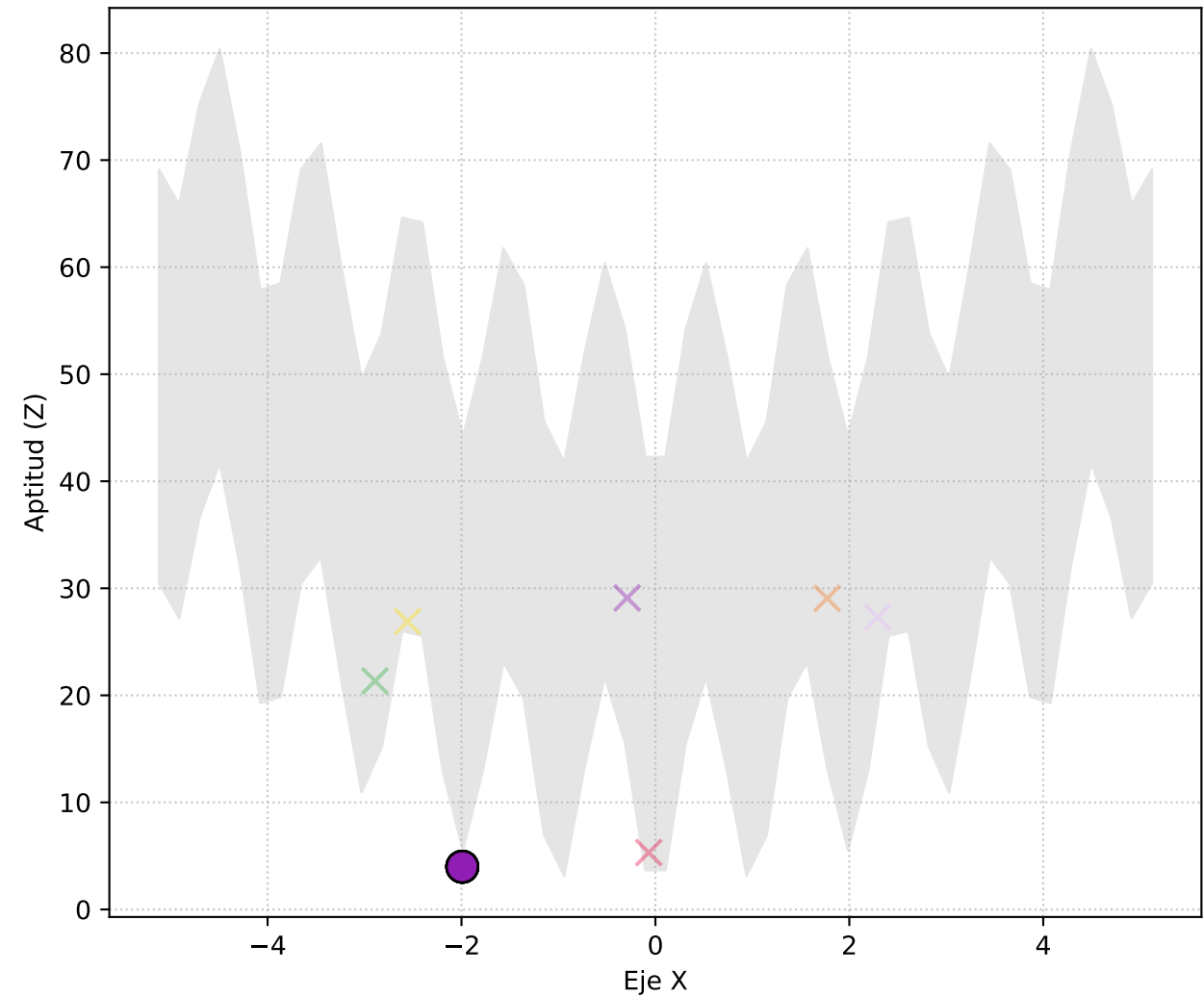
Vista Superior (X, Y)



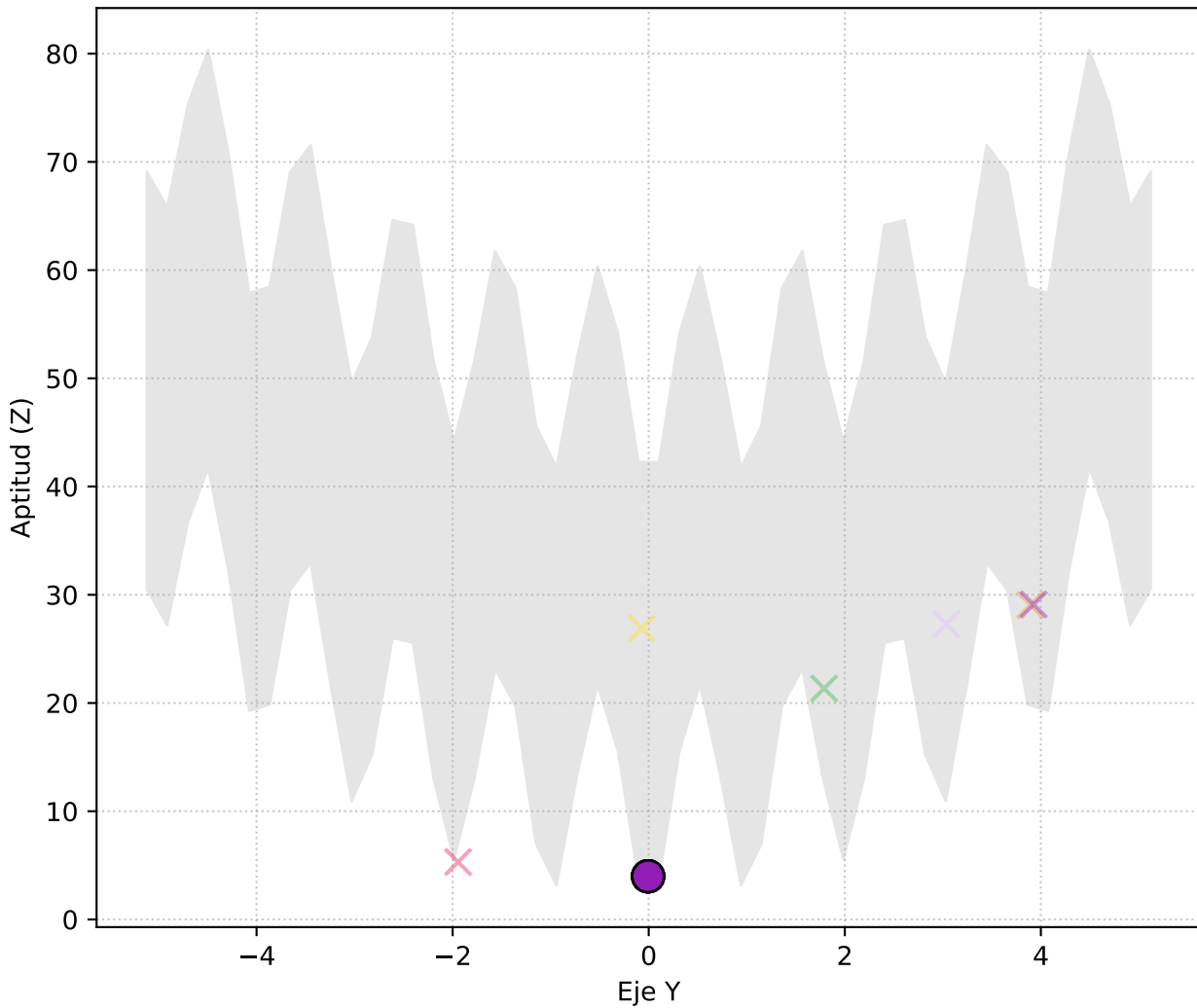
Perspectiva 3D



Vista Lateral (X vs Elevación Z)



Vista Frontal (Y vs Elevación Z)



--- GENERACIÓN 1 ---

1. Evaluación:

Bits: [0, 1, 1, 1, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0] | Dec: (-0.29, 3.93) | Aptitud: 0.0
Bits: [0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0] | Dec: (-1.76, -5.09) | Aptitud: 0.0
Bits: [1, 0, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 0, 1, 1, 1, 1, 0, 1] | Dec: (2.29, 3.03) | Aptitud: 0.04
Bits: [1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 1, 0, 1, 1] | Dec: (1.51, 3.39) | Aptitud: 0.02
Bits: [0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0] | Dec: (-3.36, 0.30) | Aptitud: 0.0
Bits: [1, 1, 0, 1, 1, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1, 0, 0, 1, 1, 0, 0] | Dec: (3.54, -2.69) | Aptitud: 0.0
Bits: [1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 0, 1] | Dec: (1.77, 3.89) | Aptitud: 0.03
Bits: [0, 1, 1, 1, 1, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 1, 1, 1, 0, 1, 1, 0, 0] | Dec: (-0.07, -1.94) | Aptitud: 0.0
Bits: [0, 0, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 1] | Dec: (-2.89, 1.79) | Aptitud: 0.0
Bits: [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1] | Dec: (-2.56, -0.07) | Aptitud: 0.0

2. Selección:

[0, 0, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 1]
[1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 0, 1]
[0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1]
[0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1]
[0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1]

3. Cruza:

Cruzando [0, 0, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 1] y [1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 1, 1] y [0, 1, 0]
Cruzando [1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 0, 1] y [0, 1, 0]
Cruzando [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 1] y [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 1]
Cruzando [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 1] y [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 1]

4. Mutación:

--- GENERACIÓN 2 ---

1. Evaluación:

Bits: [1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, 0, 0, 0, 1, 0, 1] | Dec: (2.42, 4.33) | Aptitud: 0.02
Bits: [0, 0, 1, 0, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 1] | Dec: (-3.54, 1.35) | Aptitud: 0.0
Bits: [1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1] | Dec: (1.77, 4.96) | Aptitud: 0.03
Bits: [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1] | Dec: (-2.56, -1.14) | Aptitud: 0.0
Bits: [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1] | Dec: (-2.56, -0.07) | Aptitud: 0.0
Bits: [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1] | Dec: (-2.56, -0.07) | Aptitud: 0.0
Bits: [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1] | Dec: (-2.56, -0.07) | Aptitud: 0.0
Bits: [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1] | Dec: (-2.56, -0.07) | Aptitud: 0.0
Bits: [0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1] | Dec: (-4.46, 1.87) | Aptitud: 0.0
Bits: [0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 1] | Dec: (-0.99, -0.15) | Aptitud: 0.0

2. Selección:

[0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1]
[0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1]
[0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1]
[0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1]
[0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1]

3. Cruza:

Cruzando [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1] y [0, 1, 0]
Cruzando [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1] y [0, 1, 0]
Cruzando [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1] y [0, 1, 0]
Cruzando [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1] y [0, 1, 0]

4. Mutación:

¡Mutación! [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> [0, 1, 0]
¡Mutación! [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1] -> [0, 1, 0]

--- GENERACIÓN 3 ---

1. Evaluación:

3. Cruza:

Cruzando	[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]	y	[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando	[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]	y	[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando	[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]	y	[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando	[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]	y	[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando	[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]	y	[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]

Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]

- - - GENERACIÓN 30 - - -

2. Selección:

3. Cruza:

4. Mutación:

- - - GENERACIÓN 31 - - -

2. Selección:

3. Cruza:

[illegible]

Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]

Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]

[illegible]

```

--- GENERACIÓN 85 ---
1. Evaluación:
Bits: [0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] | Dec: (-4.70, -0.01) | Aptitud: 0.00
Bits: [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] | Dec: (-2.00, -0.01) | Aptitud: 0.00
Bits: [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0] | Dec: (-2.00, -0.02) | Aptitud: 0.00
Bits: [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] | Dec: (-2.00, -0.01) | Aptitud: 0.00
Bits: [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] | Dec: (-2.00, -0.01) | Aptitud: 0.00
Bits: [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] | Dec: (-2.00, -0.01) | Aptitud: 0.00
Bits: [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] | Dec: (-2.00, -0.01) | Aptitud: 0.00
Bits: [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] | Dec: (-2.00, -0.01) | Aptitud: 0.00
Bits: [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] | Dec: (-2.00, -0.01) | Aptitud: 0.00
2. Selección:
[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]

```


Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1]

- - - GENERACIÓN 90 - - -

2. Selección:

3. Cruza:

4. Mutación:

2. Selección:

3. Cruza:

Bits: [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] | Dec: (-1.99, -0.01) | Aptitud: 0.
Bits: [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] | Dec: (-1.99, -0.01) | Aptitud: 0.
Bits: [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] | Dec: (-1.99, -0.01) | Aptitud: 0.
Bits: [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] | Dec: (-1.99, -0.01) | Aptitud: 0.

2. Selección:

[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
[0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]

3. Cruza:

Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
Cruzando [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] y [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]

4. Mutación:

¡Mutación! [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1] -> [0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1]

===== RESULTADO FINAL =====

Bits: [0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1]
Valor (X, Y): (-1.99, -0.01)
Aptitud Alcanzada: 0.20